

COMPUTER NETWORKS

(25MC410)

I M.C.A II Semester

2025-26 EVEN SEMESTER

Name of the Faculty : Dr. K. Santhi Sri
Associate Professor, Department of CA
VFSTR (Deemed to be University)

Syllabus

Module-2

UNIT-2

18L+18T+0P+18SL=54 Hours

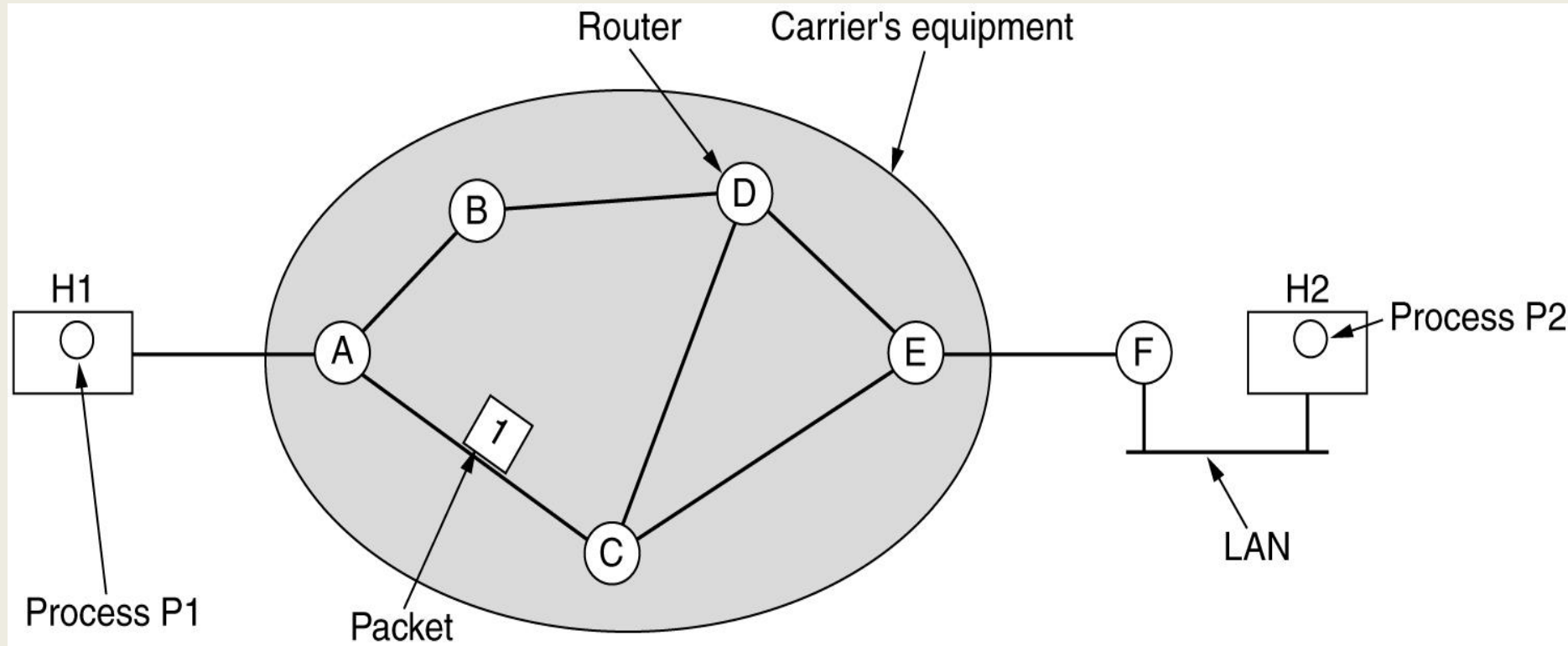
THE NETWORK LAYER

Network Layer Design Issues, Routing Algorithms in A Single Network, Traffic Management At The Network Layer, Quality of Service and Application QOE, Internetworking, The Network Layer in the Internet (IPv4, IPv6, ICMP, ARP).

Network Layer Design Issues

- Store-and-Forward Packet Switching
- Services Provided to the Transport Layer
- Implementation of Connectionless Service
- Implementation of Connection-Oriented Service
- Comparison of Virtual-Circuit and Datagram Subnets

Store-and-Forward Packet Switching

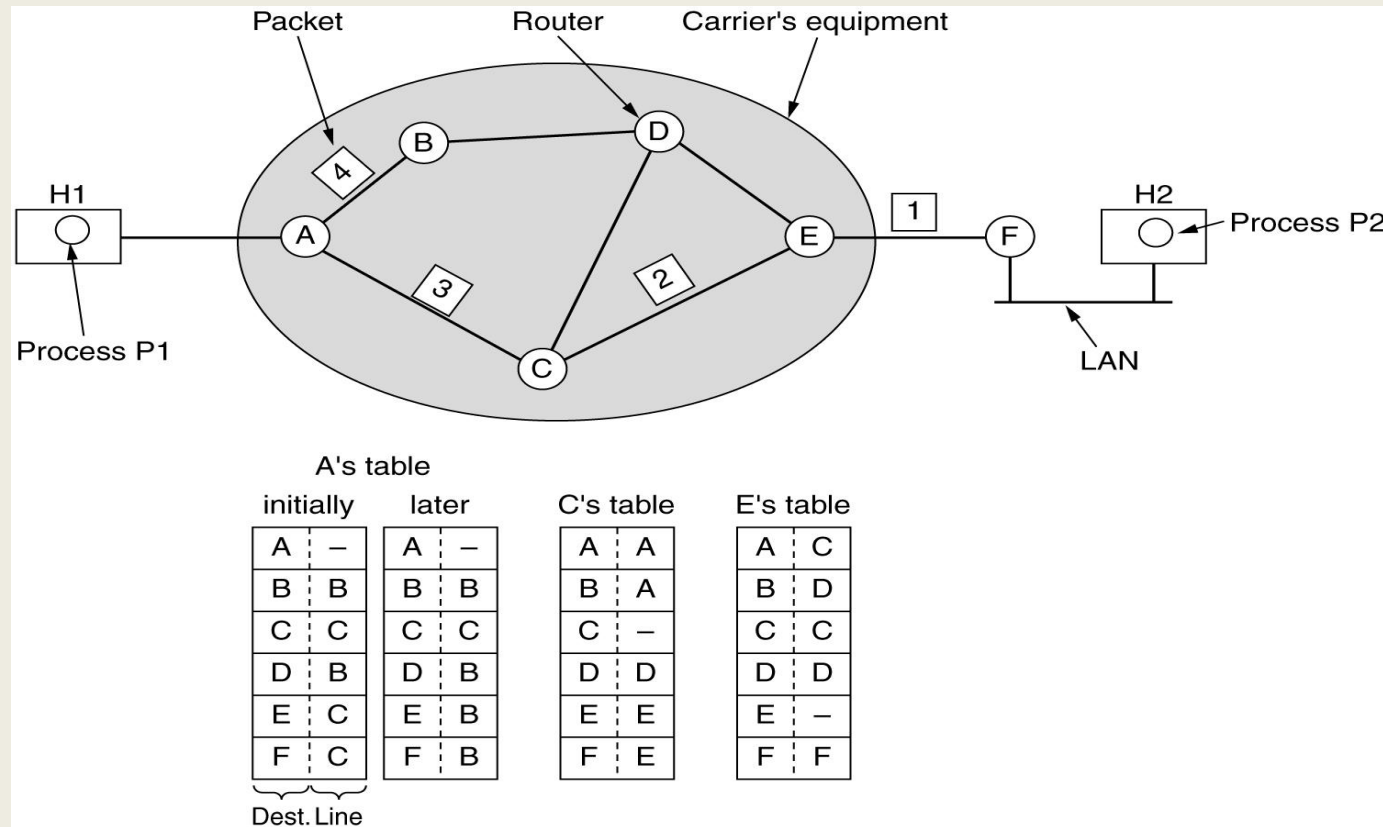


The environment of the network layer protocols.

Services provided to the Transport Layer

- The services should be independent of the router technology
- The transport layer should be shielded from the number, type and the topology of the routers present.
- The network addresses made available to the transport layer should use a uniform numbering plan, even across LANs and WANs.

Implementation of Connectionless Service



Routing within a datagram subnet.

Implementation of Connectionless Service

- If connectionless service is offered, packets are injected into the network individually and routed independently of each other.
- No advance setup is needed.
- In this context, the packets are frequently called **datagrams** (in analogy with telegrams) and the network is called a **datagram network**.

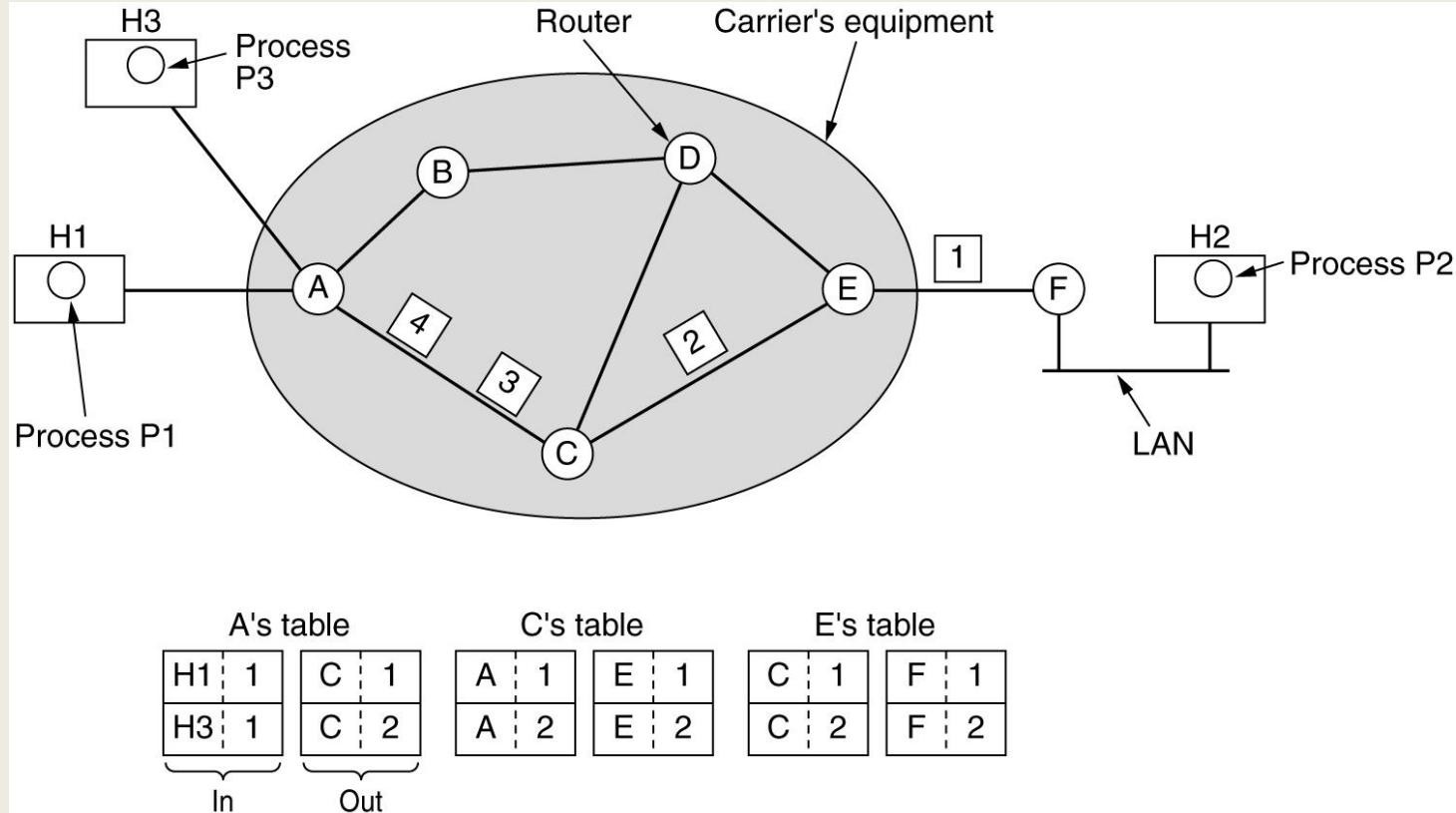
Implementation of connection-oriented Service

- If connection-oriented service is used, a path from the source router all the way to the destination router must be established before any data packets can be sent.
- This connection is called a **VC (virtual circuit)**, in analogy with the physical circuits set up by the telephone system, and the network is called a **virtual-circuit network**.
- The idea behind virtual circuits is to avoid having to choose a new route for every packet sent.

Implementation of connection-oriented Service

- When a connection is established, a route from the source machine to the destination machine is chosen as part of the connection setup and stored in tables inside the routers.
- That route is used for all traffic flowing over the connection, exactly the same way that the telephone system works.
- When the connection is released, the virtual circuit is also terminated.
- With connection-oriented service, each packet carries an identifier telling which virtual circuit it belongs to.

Implementation of Connection-Oriented Service



Routing within a virtual-circuit subnet.

Comparison of Virtual-Circuit and Datagram Subnets

Issue	Datagram subnet	Virtual-circuit subnet
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Routing Algorithms

- Routing algorithm is the part of the network layer software responsible for deciding which output line an incoming packet should be transmitted on.
- If the subnet uses datagram internally, this decision must be made a new for every arriving data packet.
- If the subnet uses virtual circuit internally, routing decisions are made only when a new virtual circuit is being setup (**session routing** because a route remains in force for an entire session)

Routing Algorithms

- Router has two processes inside it.
- One of them handles each packet as it arrives, looking up the outgoing line to use for it in the routing tables. This process is **forwarding**.
- The other process is responsible for filling in and updating the routing tables.
- Desirable properties of routing algorithm
 - Correctness, simplicity, robustness, stability, fairness and optimality.

Routing Algorithms

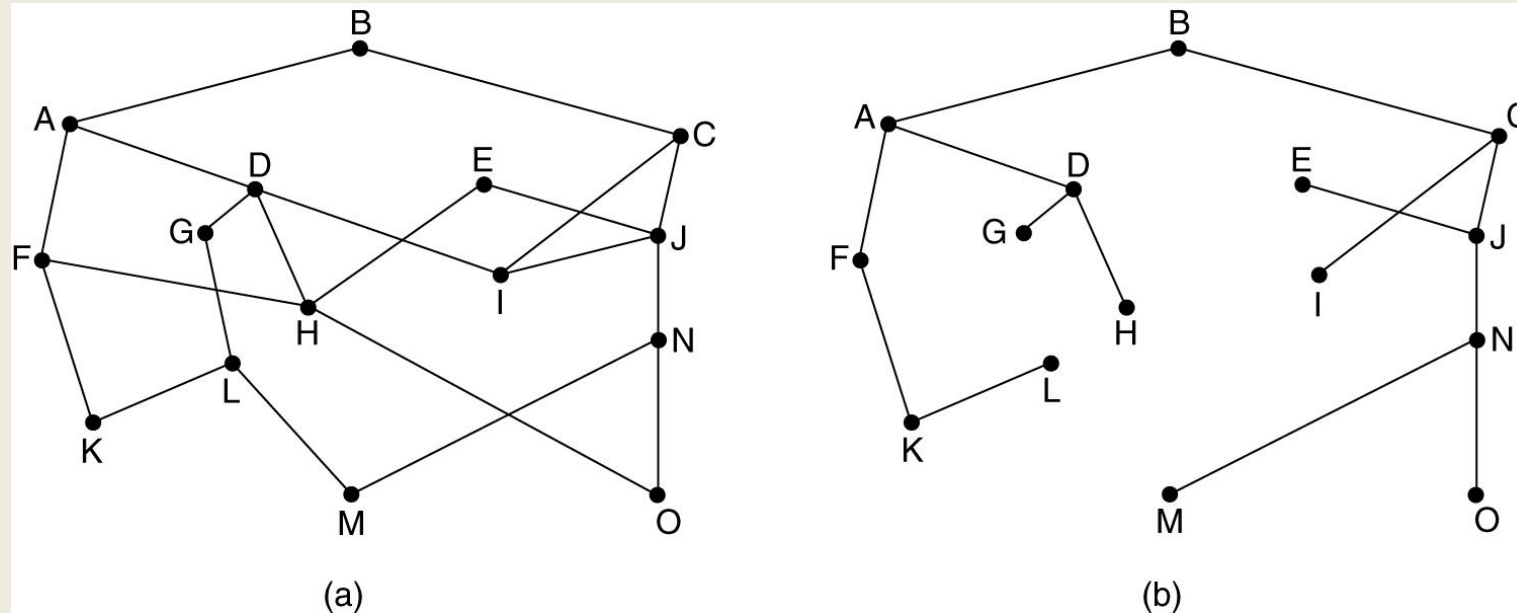
Routing algorithms grouped into two major classes.

- **Nonadaptive algorithms:** do not base their routing decisions on any measurements or estimates of the current topology and traffic. This procedure is sometimes called **static routing**. Because it does not respond to failures, static routing is mostly useful for situations in which the routing choice is clear.
- **Adaptive algorithms:** change their routing decisions to reflect changes in the topology, and sometimes changes in the traffic as well. These **dynamic routing** algorithms differ in where they get their information (e.g., locally, from adjacent routers, or from all routers), when they change the routes (e.g., when the topology changes, or every ΔT seconds as the load changes) and what metric is used for optimization (e.g., distance, number of hops, or estimated transit time)

The Optimality Principle

- It states that if router J is on the optimal path from router I to router K , then the optimal path from J to K also falls along the same route.
- To see this, call the part of the route from I to J $r1$ and the rest of the route $r2$. If a route better than $r2$ existed from J to K , it could be concatenated with $r1$ to improve the route from I to K , contradicting our statement that $r1r2$ is optimal.
- The set of optimal routes from all sources to a given destination form a tree rooted at the destination. Such a tree is called a **sink tree**
- **Sink tree:** The set of optimal routes from all sources to the given destination form a tree rooted at the destination. Such a tree is called a sink tree.

The Optimality Principle



(a) A subnet. (b) A sink tree for router B.

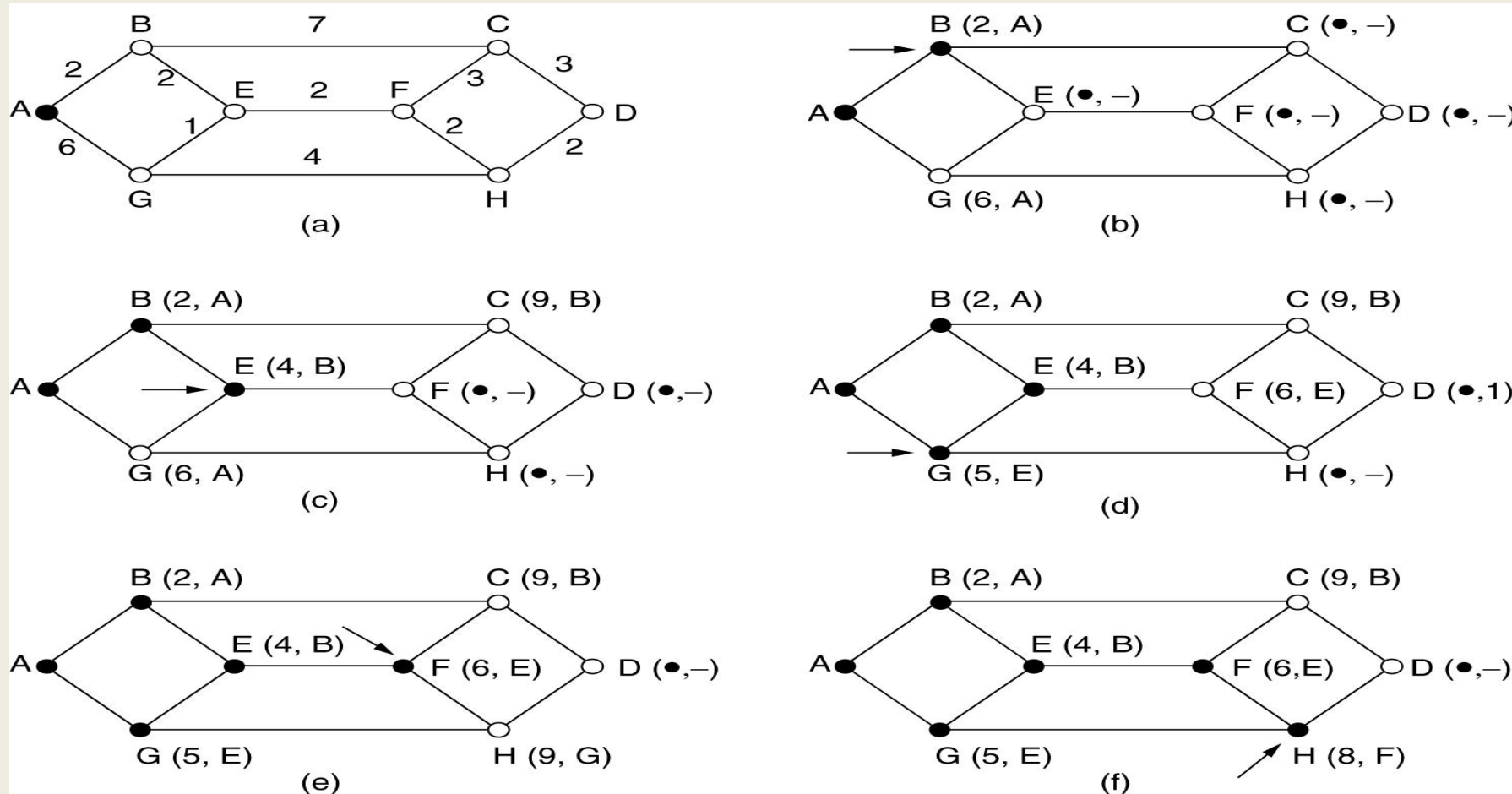
Shortest Path Routing

- The idea is to build a graph of the network, with each node of the graph representing a router and each edge of the graph representing a communication line, or link.
- To choose a route between a given pair of routers, the algorithm just finds the shortest path between them on the graph.
- One way of measuring path length is the number of hops. Using this metric, the paths *ABC* and *ABE* are equally long. Another metric is the geographic distance in kilometers, in which case *ABC* is clearly much longer than *ABE*.

Shortest Path Routing

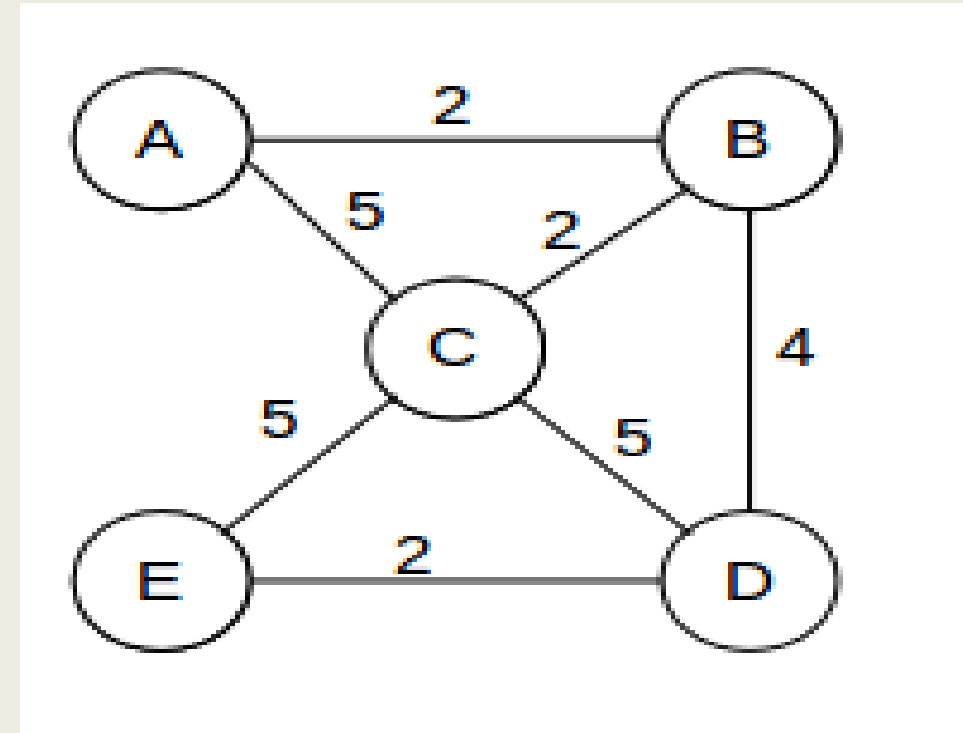
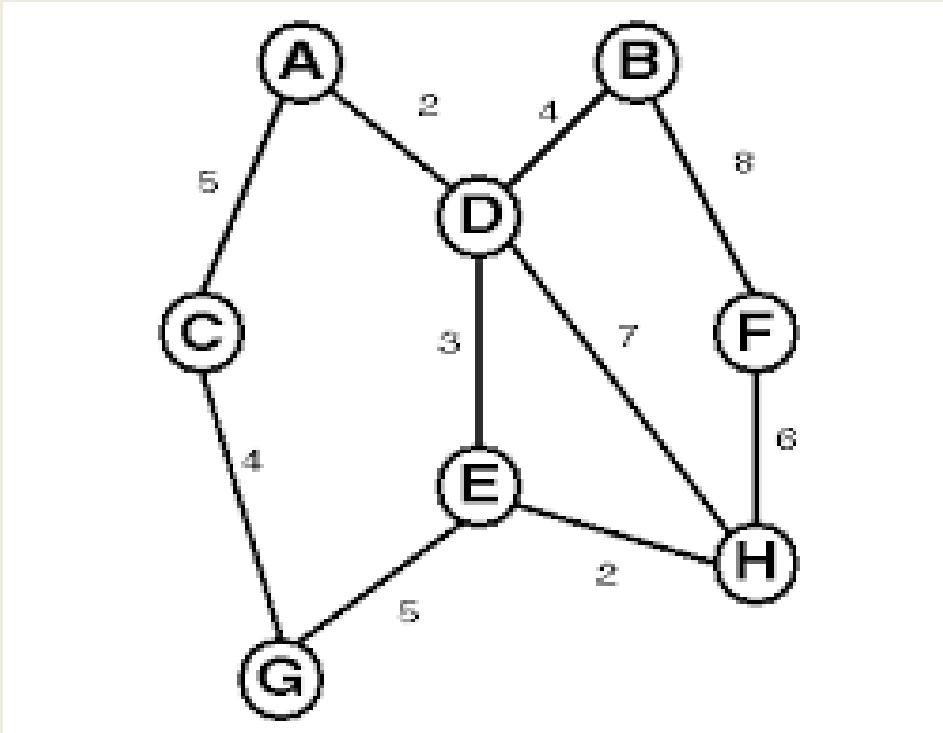
1. We maintain a pair of cost of the path so far accessed & the predecessor in the path at each node.
2. Initially cost of the path is set to infinity and predecessor is empty.
3. As the algorithm progresses, the labels may change.
4. At any stage, when it is clear that the current label is shortest path b/w a node & the source node, then it is marked as permanent.
5. The algorithm terminates when the destination node gets a permanent label.

Shortest Path Routing

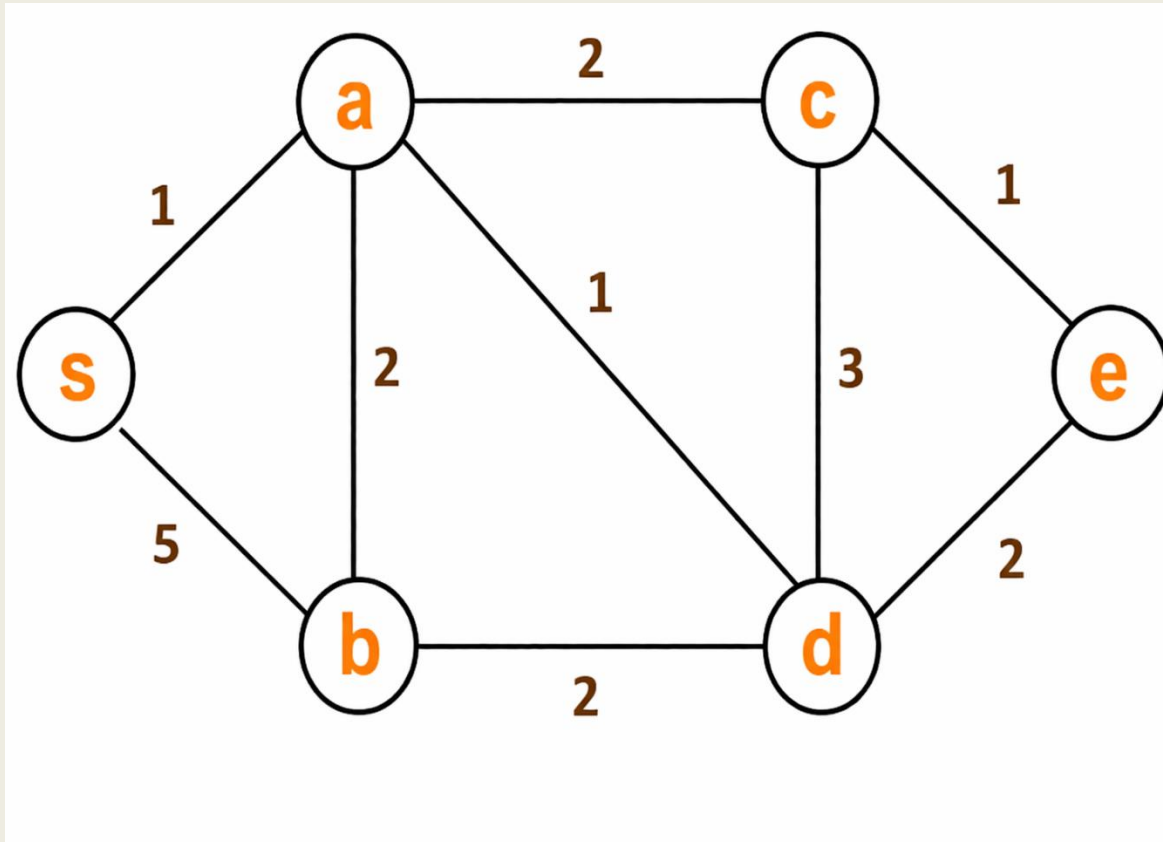


The first 5 steps used in computing the shortest path from A to D.
 The arrows indicate the working node.

Shortest Path Routing



Shortest Path Routing



Flooding

- Every incoming packet is sent out on every outgoing line except the one it arrived on.
- Drawback of it is, it generates vast number of duplicate packets.
- One measure is to have a hop counter contained in the header of each packet that is decremented at each hop, with the packet being discarded when the counter reaches zero.
- Ideally, the hop counter should be initialized to the length of the path from source to destination. If the sender does not know how long the path is, it can initialize the counter to the worst case, namely, the full diameter of the network.

- Flooding with a hop count can produce an exponential number of duplicate packets as the hop count grows and routers duplicate packets they have seen before.
- A better technique for damming the flood is to have routers keep track of which packets have been flooded, to avoid sending them out a second time.
- One way to achieve this goal is to have the source router put a sequence number in each packet it receives from its hosts. Each router then needs a list per source router telling which sequence numbers originating at that source have already been seen. If an incoming packet is on the list, it is not flooded.

Distance Vector Routing

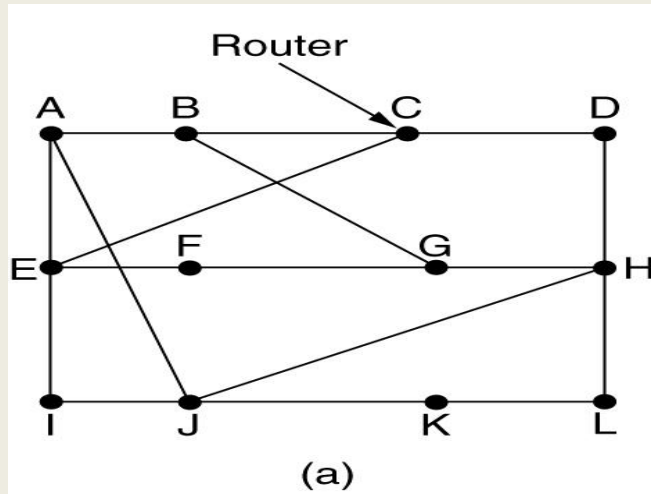
- It is a dynamic routing algorithm, which takes current network traffic in account.
- It operates by having each router maintain a table giving the best known distance to each destination and which line to use to get there. These tables are updated by exchanging information with the neighbors. Eventually, every router knows the best link to reach each destination.
- In distance vector routing, each router maintains a routing table indexed by, and containing one entry for each router in the network.

- Each entry in routing table contains 2 parts: the preferred outgoing line and an estimate of the time or distance to that destination.
- The router is assumed to know the “distance” to each of its neighbors. If the metric is hops, the distance is just one hop.
- If the metric is propagation delay, the router can measure it directly with special ECHO packets that the receiver just timestamps and sends back as fast as it can.

Distance Vector Routing

- Assume that delay is used as a metric and that the router knows the delay to each of its neighbors.
- Once every T msec, each router sends to each neighbor a list of its estimated delays to each destination. It also receives a similar list from each neighbor.
- Imagine that one of these tables has just come in from neighbor X , with X_i being X 's estimate of how long it takes to get to router i . If the router knows that the delay to X is m msec, it also knows that it can reach router i via X in $X_i + m$ msec.
- By performing this calculation for each neighbor, a router can find out which estimate seems the best and use that estimate and the corresponding link in its new routing table.

Distance Vector Routing:



To	A	I	H	K	New estimated delay from J	
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	—
K	24	22	22	0	6	K
L	29	33	9	9	15	K

JA delay is 8	JI delay is 10	JH delay is 12	JK delay is 6
---------------	----------------	----------------	---------------

Vectors received from J's four neighbors

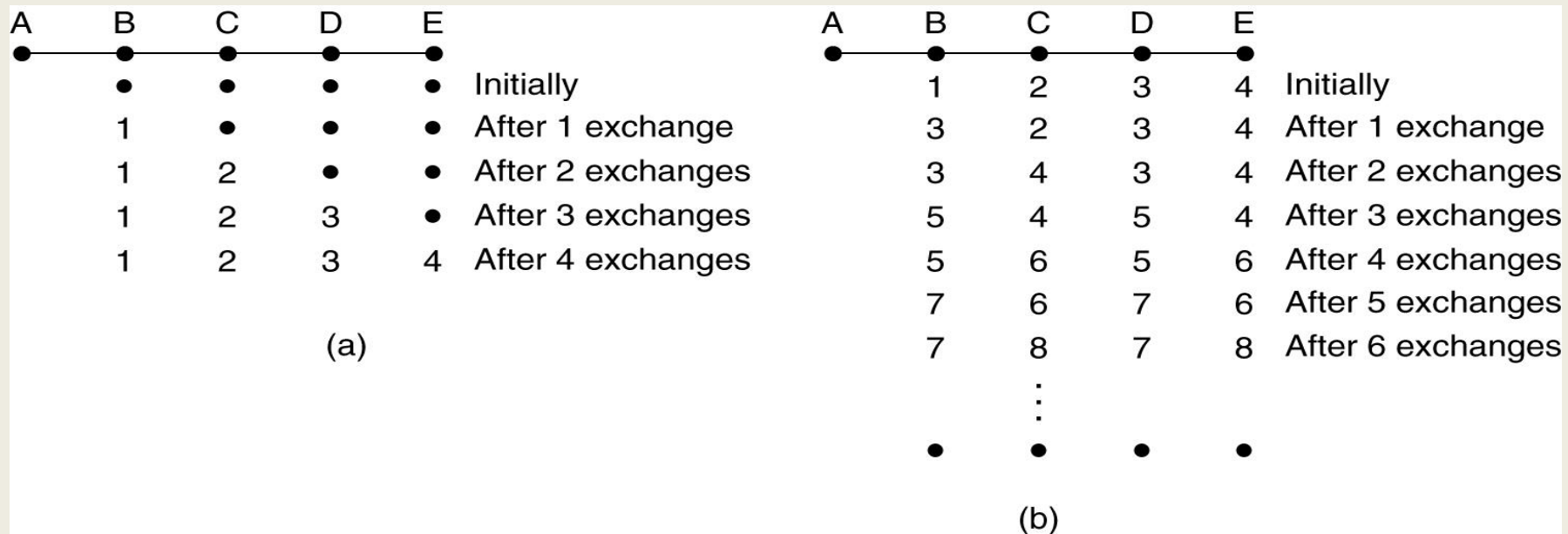
New routing table for J	
Line	
8	A
20	A
28	I
20	H
17	I
30	I
18	H
12	H
10	I
0	—
6	K
15	K

(b)

(a) A subnet. (b) Input from A, I, H, K, and the new routing table for J.

Distance Vector Routing:

- In particular, it reacts rapidly to good news, but leisurely to bad news.
- Consider a router whose best route to destination X is long. If, on the next exchange, neighbor A suddenly reports a short delay to X , the router just switches over to using the line to A to send traffic to X . In one vector exchange, the good news is processed.



The count-to-infinity problem.

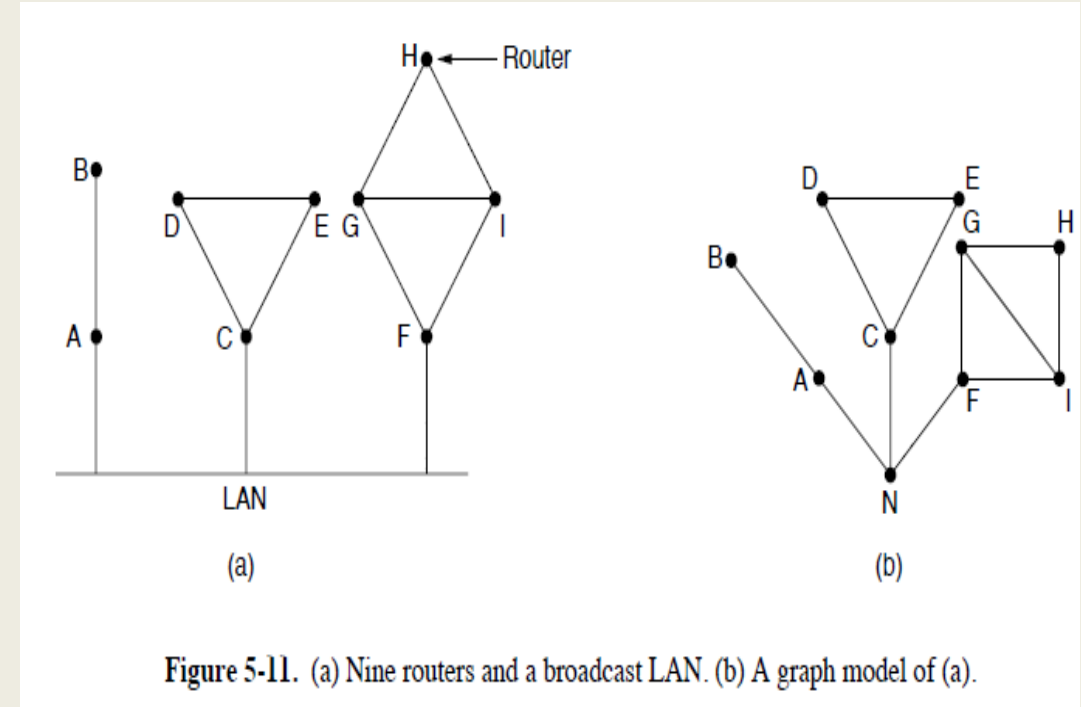
Link State Routing:

- The idea behind link state routing is fairly simple and can be stated as five parts.
- Each router must do the following things to make it work:
 1. Discover its neighbors and learn their network addresses.
 2. Set the distance or cost metric to each of its neighbors.
 3. Construct a packet telling all it has just learned.
 4. Send this packet to and receive packets from all other routers.
 5. Compute the shortest path to every other router.

Link State Routing:

Learning about the Neighbors:

- When a router is booted, its first task is to learn who its neighbors are.
- It accomplishes this goal by sending a special HELLO packet on each point-to-point line.
- The router on the other end is expected to send back a reply giving its name.
- These names must be globally unique.



Link State Routing:

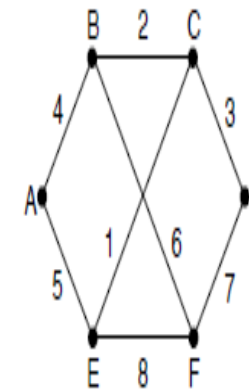
Setting Link Costs:

- The link state routing algorithm requires each link to have a distance or cost metric for finding shortest paths.
- The cost to reach neighbors can be set automatically, or configured by the network operator.
- The most direct way to determine this delay is to send over the line a special ECHO packet that the other side is required to send back immediately.
- By measuring the roundtrip time and dividing it by two, the sending router can get an estimate of the delay.

Link State Routing:

Building Link State Packets:

- The packet starts with the identity of the sender, followed by a sequence number and age and a list of neighbors.
- The cost to each neighbor is also given.
- Building the link state packets is easy.
- The hard part is determining when to build them.
- One possibility is to build them periodically, at regular intervals.
- Another possibility is to build them when some specific event occurs, such as a line or neighbor going down or coming back up again or changing its properties.



(a)

Link		State		Packets	
A	B	C	D	E	F
Seq.	Seq.	Seq.	Seq.	Seq.	Seq.
Age	Age	Age	Age	Age	Age
B 4	A 4	B 2	C 3	A 5	B 6
E 5	C 2	D 3	F 7	C 1	D 7
	F 6	E 1		F 8	E 8

(b)

Figure 5-12. (a) A network. (b) The link state packets for this network.

Link State Routing:

Distributing the Link State Packets:

- All of the routers must get all of the link state packets quickly and reliably.
- The fundamental idea is to use flooding to distribute the link state packets to all routers.
- To keep the flood in check, each packet contains a sequence number that is incremented for each new packet sent.
- Routers keep track of all the (source router, sequence) pairs they see.
- When a new link state packet comes in, it is checked against the list of packets already seen.
- If it is new, it is forwarded on all lines except the one it arrived on.
- If it is a duplicate, it is discarded.
- If a packet with a sequence number lower than the highest one seen so far ever arrives, it is rejected as being obsolete as the router has more recent data.

Problems with the previous algorithm and solutions:

- First, if the sequence numbers wrap around, confusion will reign.
- The solution here is to use a 32-bit sequence number. With one link state packet per second, it would take 137 years to wrap around.
- Second, if a router ever crashes, it will lose track of its sequence number.
- If it starts again at 0, the next packet it sends will be rejected as a duplicate.
- Third, if a sequence number is ever corrupted and 65,540 is received instead of 4 (a 1-bit error), packets 5 through 65,540 will be rejected as obsolete, since the current sequence number will be thought to be 65,540.

Problems with the previous algorithm and solutions:

- The solution to these problems is to include the age of each packet after the sequence number and decrement it once a second.
- When the age hits zero, the information from that router is discarded.
- The *Age* field is also decremented by each router during the initial flooding process, to make sure no packet can get lost and live for an indefinite period of time (a packet with age zero is discarded).
- When a link state packet comes in to a router for flooding, it is not queued for transmission immediately.
- Instead, it is put in a holding area to wait a short while in case more links are coming up or going down.
- If another link state packet from the same source comes in before the first packet is transmitted, their sequence numbers are compared.
- If they are equal, the duplicate is discarded. If they are different, the older one is thrown.

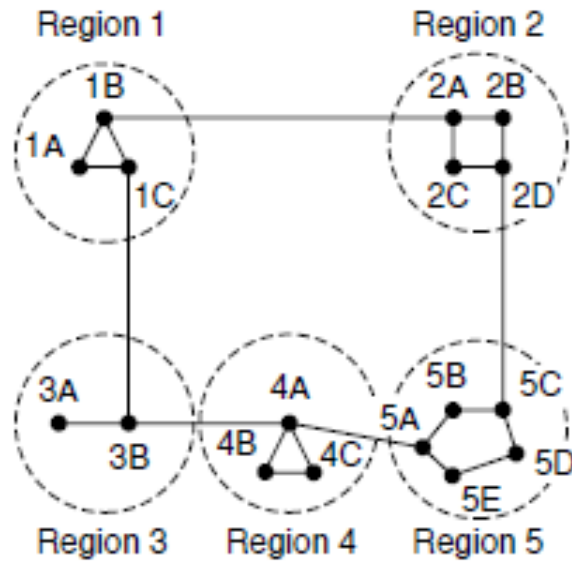
Computing the New Routes:

- Once a router has accumulated a full set of link state packets, it can construct the entire network graph because every link is represented.
- Every link is, in fact, represented twice, once for each direction.
- The different directions may even have different costs.
- The shortest-path computations may then find different paths from router *A* to *B* than from router *B* to *A*.
- Now Dijkstra's algorithm can be run locally to construct the shortest paths to all possible destinations.
- The results of this algorithm tell the router which link to use to reach each destination. This information is installed in the routing tables, and normal operation is resumed.

Hierarchical Routing within a Network:

- As networks grow in size, the router routing tables grow proportionally.
- Not only is router memory consumed by ever-increasing tables, but more CPU time is needed to scan them and more bandwidth is needed to send status reports about them.
- At a certain point, the network may grow to a size where it is no longer feasible for every router to have an entry for every other router, so the routing will have to be done hierarchically, through the use of routing areas.
- When hierarchical routing is used, the routers are divided into what we will call regions or areas.
- Each router knows all the details about how to route packets to destinations within its own region but knows nothing about the internal structure of other regions.
- When different networks are interconnected, it is natural to regard each one as a separate region to free the routers in one network from having to know the topological structure of the other ones.

Hierarchical Routing within a Network:



(a)

Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

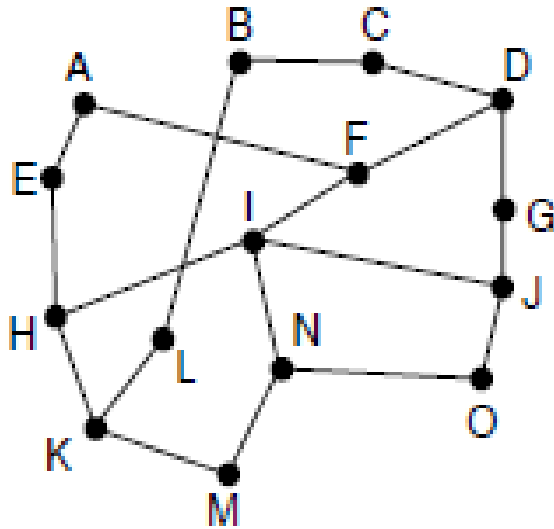
(c)

Figure 5-14. Hierarchical routing.

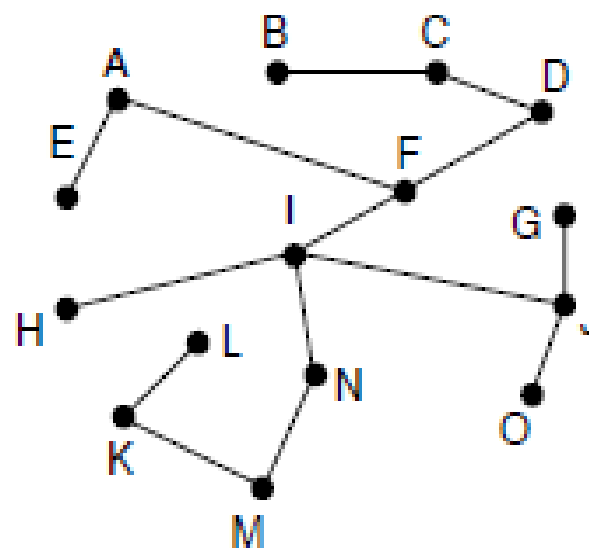
- In some applications, hosts need to send messages to many or all other hosts.
- Sending a packet to all destinations simultaneously is called broadcasting.
- One broadcasting method that requires no special features from the network is for the source to simply send a distinct packet to each destination.
- An improvement is **multidestination routing**, in which each packet contains either a list of destinations or a bit map indicating the desired destinations.
- When a packet arrives at a router, the router checks all the destinations to determine the set of output lines that will be needed.
- The router generates a new copy of the packet for each output line to be used and includes in each packet only those destinations that are to use the line.
- In effect, the destination set is partitioned among the output lines. After a sufficient number of hops, each packet will carry only one destination like a normal packet.

- Better broadcast routing technique: flooding. When implemented with a sequence number per source, flooding uses links efficiently with a decision rule at routers that is relatively simple.
- The idea for **re verse path forwarding** is elegant and remarkably simple.
- When a broadcast packet arrives at a router, the router checks to see if the packet arrived on the link that is normally used for sending packets *toward* the source of the broadcast.
- If so, there is an excellent chance that the broadcast packet itself followed the best route from the router and is therefore the first copy to arrive at the router.
- The router forwards copies of it onto all links except the one it arrived on.
- If, however, the broadcast packet arrived on a link other than the preferred one for reaching the source, the packet is discarded as a likely duplicate.

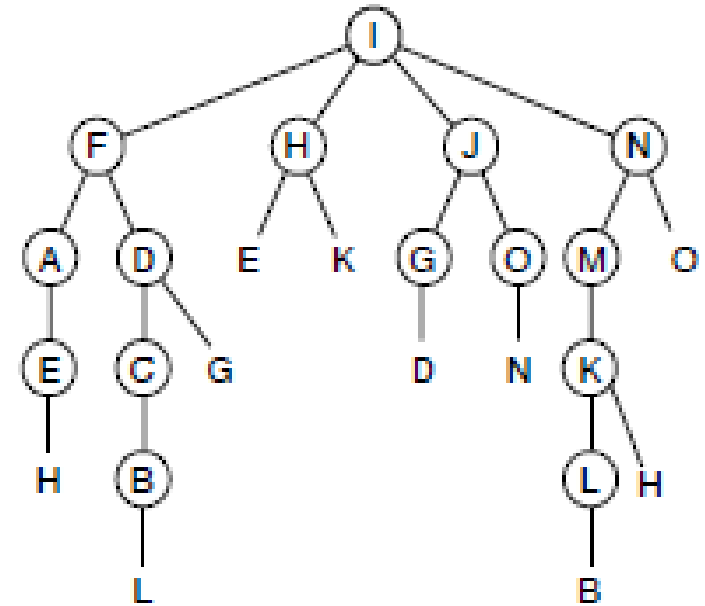
Broadcast Routing:



(a)



(b)

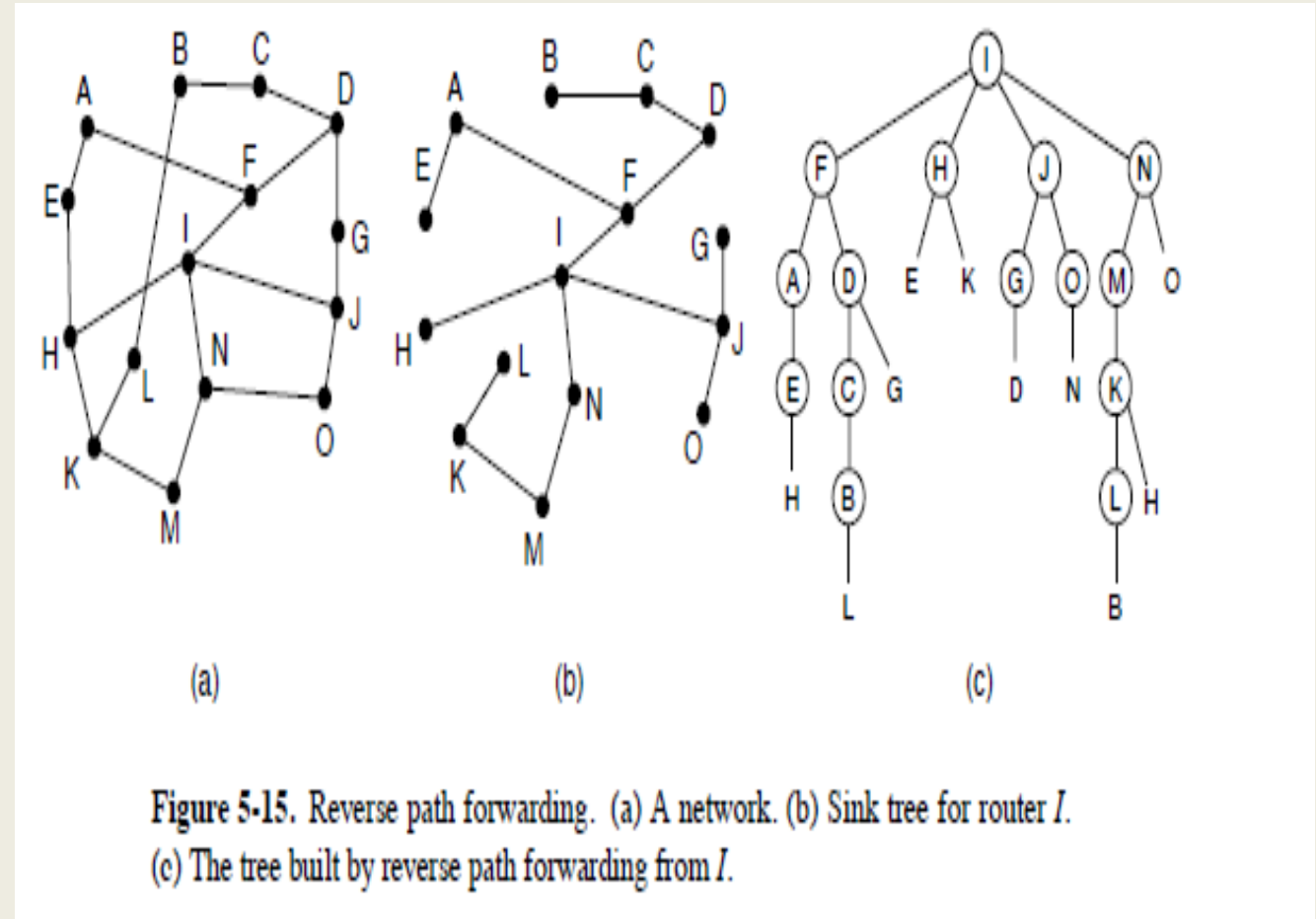


(c)

Figure 5-15. Reverse path forwarding. (a) A network. (b) Sink tree for router *I*. (c) The tree built by reverse path forwarding from *I*.

Broadcast Routing:

- On the first hop, I sends packets to F, H, J, and N, as indicated by the second row of the tree.
- Each of these packets arrives on the preferred path to I and is so indicated by a circle around the letter.
- On the second hop, eight packets are generated, two by each of the routers that received a packet on the first hop.
- As it turns out, all eight of these arrive at previously unvisited routers, and five of these arrive along the preferred line.
- Of the six packets generated on the third hop, only three arrive on the preferred path (at C, E, and K); the others are duplicates.
- After five hops and 24 packets, the broadcasting terminates, compared with four hops and 14 packets had the sink tree been followed exactly.



- Our last broadcast algorithm improves on the behavior of reverse path forwarding.
- It makes explicit use of the sink tree—or any other convenient spanning tree for that matter—for the router initiating the broadcast.
- A spanning tree is a subset of the network that includes all the routers but contains no loops. Sink trees are spanning trees.
- If each router knows which of its lines belong to the spanning tree, it can copy an incoming broadcast packet onto all the spanning tree lines except the one it arrived on.
- This method makes excellent use of bandwidth, generating the absolute minimum number of packets necessary to do the job.
- In Fig. 5-15, for example, when the sink tree of part (b) is used as the spanning tree, the broadcast packet is sent with the minimum 14 packets.
- The only problem is that each router must have knowledge of some spanning tree for the method to be applicable.

Multicast Routing:

- Some applications, such as a multiplayer game or live video of a sports event streamed to many viewing locations, send packets to multiple receivers.
- Sending a message to such a group is called multicasting, and the routing algorithm used is called multicast routing.
- All multicasting schemes require some way to create and destroy groups and to identify which routers are members of a group.
- we will assume that each group is identified by a multicast address and that routers know the groups to which they belong.

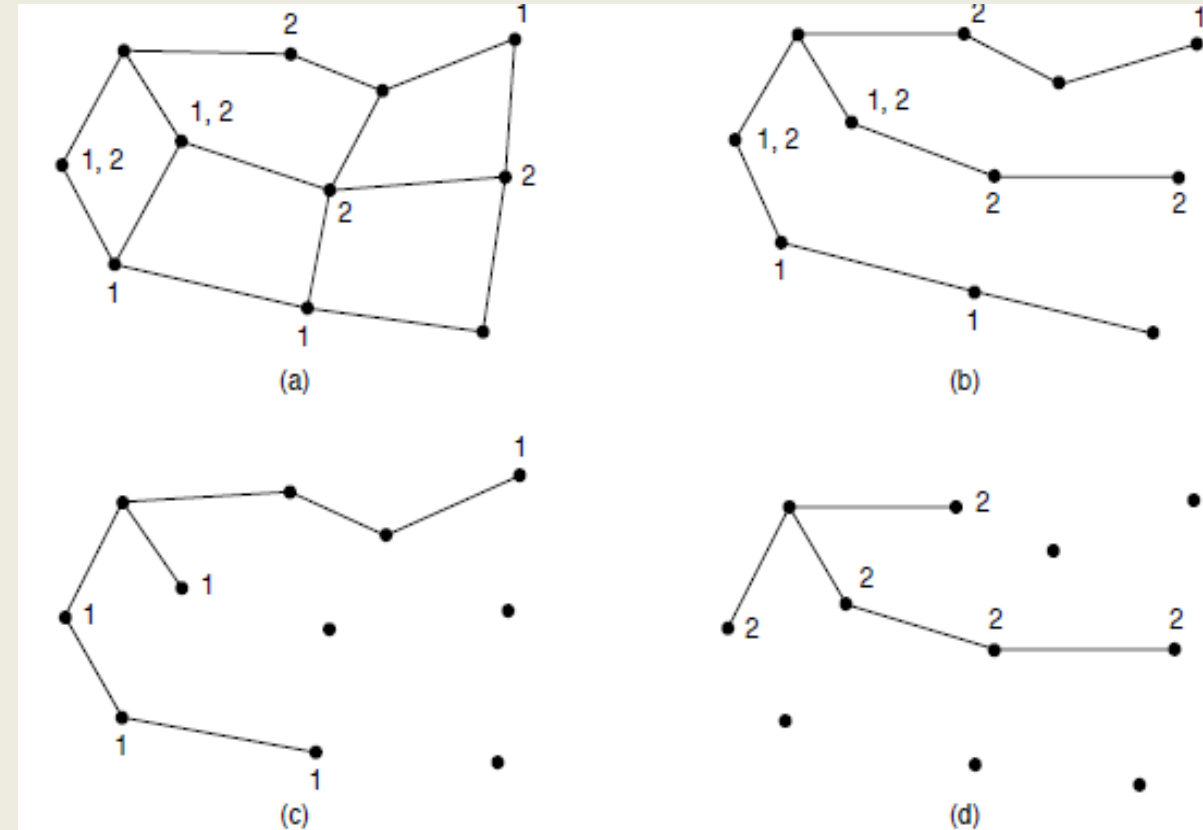
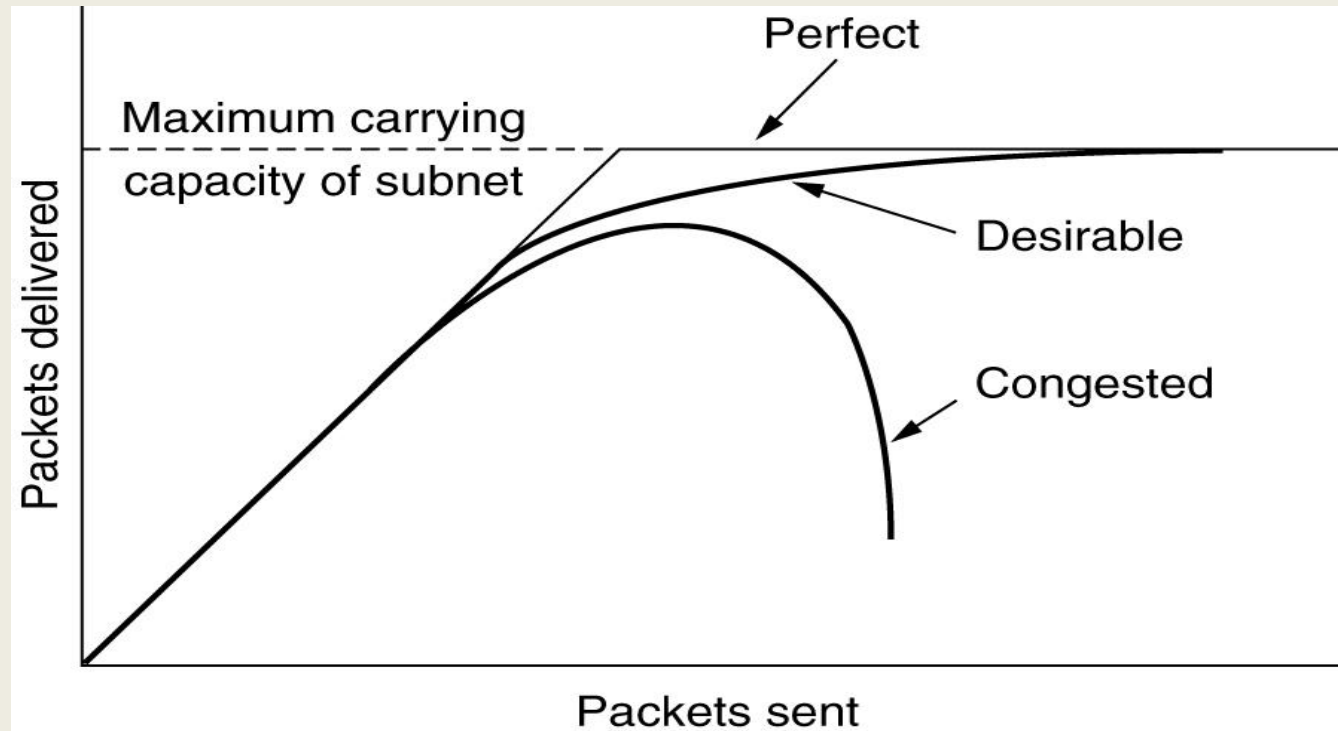


Figure 5-16. (a) A network. (b) A spanning tree for the leftmost router. (c) A multicast tree for group 1. (d) A multicast tree for group 2.

Congestion Control Algorithms

- Too many packets present in (a part of) the network causes packet delay and loss that degrades performance. This situation is called **congestion**.
- The network and transport layers share the responsibility for handling congestion. Since congestion occurs within the network, it is the network layer that directly experiences it and must ultimately determine what to do with the excess packets.
- The most effective way to control congestion is to reduce the load that the transport layer is placing on the network. This requires the network and transport layers to work together.

Congestion



When too much traffic is offered, congestion sets in and performance degrades sharply.

Factors to cause congestion

- If all of a sudden, streams of packet begin arriving on the 3or 4 input lines and all need the same output line.
- If there is insufficient memory to hold all of them, packets will be lost. Adding more memory may help up to a point, but Nagle (1987) realized that if routers have an infinite amount of memory, congestion gets worse, not better. This is because by the time packets get to the front of the queue, they have already timed out (repeatedly) and duplicates have been sent.
- Low-bandwidth links or routers that process packets more slowly than the line rate can also become congested.

Approaches to Congestion Control

- Two solutions come to mind: increase the resources or decrease the load.
- These solutions are usually applied on different time scales to either prevent congestion or react to it once it has occurred.

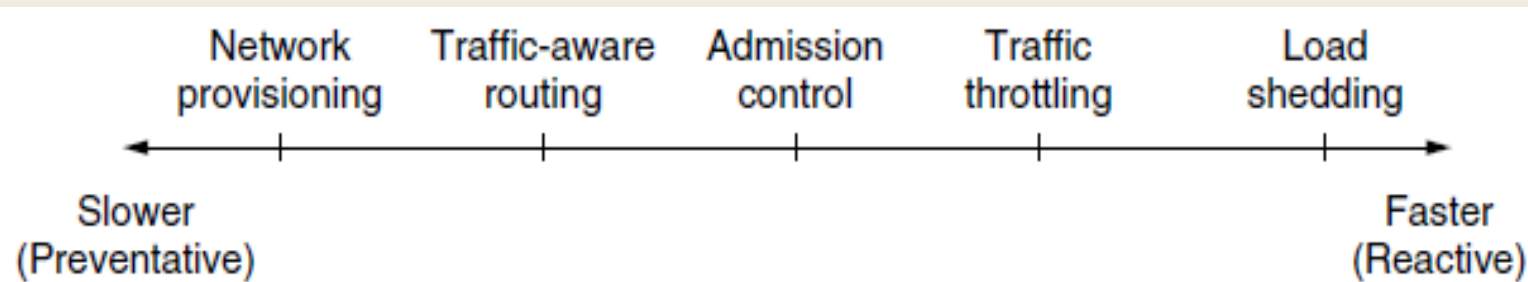


Figure 5-22. Timescales of approaches to congestion control.

Approaches to Congestion Control

- The most basic way to avoid congestion is to build a network that is well matched to the traffic that it carries. If there is a low-bandwidth link on the path along which most traffic is directed, congestion is likely.
- Sometimes resources can be added dynamically when there is serious congestion. This is called **provisioning** and happens on a time scale of months, driven by long-term traffic trends.
- To make the most of the existing network capacity, routes can be tailored to traffic patterns that change during the day as network users wake and sleep in different time zones. This is called **traffic-aware routing**.

Approaches to Congestion Control

- Sometimes it is not possible to increase capacity. The only way then to beat back the congestion is to decrease the load. In a virtual-circuit network, new connections can be refused if they would cause the network to become congested. This is called **admission control**.
- At a finer granularity, when congestion is imminent the network can deliver feedback to the sources whose traffic flows are responsible for the problem. The network can request these sources to throttle their traffic, or it can slow down the traffic itself.

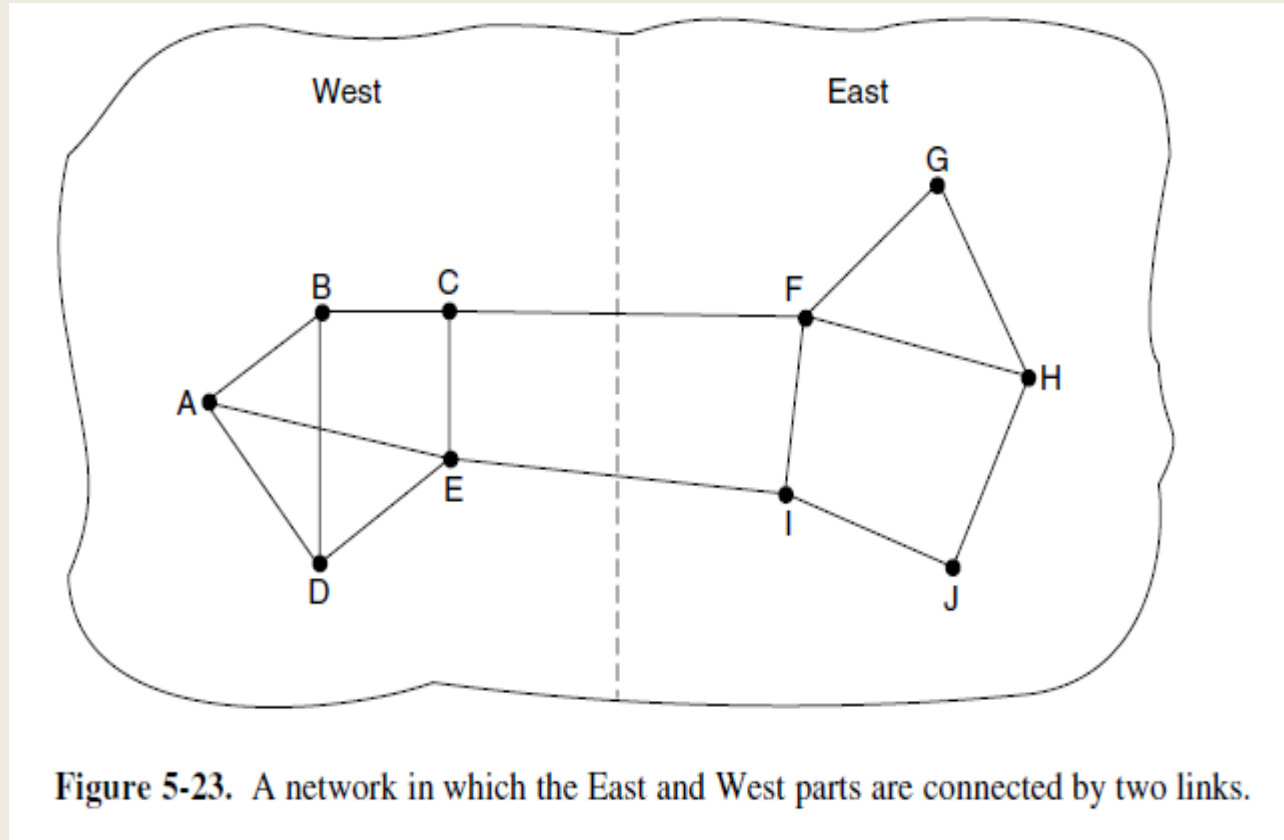
Traffic-Aware Routing

- The routing schemes we looked used fixed link weights. These schemes adapted to changes in topology, but not to changes in load.
- The goal in taking load into account when computing routes is to shift traffic away from hotspots that will be the first places in the network to experience congestion.
- The most direct way to do this is to set the link weight to be a function of the (fixed) link bandwidth and propagation delay plus the (variable) measured load or average queuing delay. Least-weight paths will then favor paths that are more lightly loaded, all else being equal.

Traffic-Aware Routing

- Consider the network which is divided into two parts, East and West, connected by two links, *CF* and *EI*.
- Suppose that most of the traffic between East and West is using link *CF*, and, as a result, this link is heavily loaded with long delays.
- Including queueing delay in the weight used for the shortest path calculation will make *EI* more attractive. After the new routing tables have been installed, most of the East-West traffic will now go over *EI*, loading this link.
- Consequently, in the next update, *CF* will appear to be the shortest path. As a result, the routing tables may oscillate wildly, leading to erratic routing and many potential problems.

Traffic-Aware Routing



Traffic-Aware Routing

- If load is ignored and only bandwidth and propagation delay are considered, this problem does not occur. Attempts to include load but change weights within a narrow range only slow down routing oscillations.
- Two techniques can contribute to a successful solution.
- The first is multipath routing, in which there can be multiple paths from a source to a destination. In our example this means that the traffic can be spread across both of the East to West links.
- The second one is for the routing scheme to shift traffic across routes slowly enough that it is able to converge.

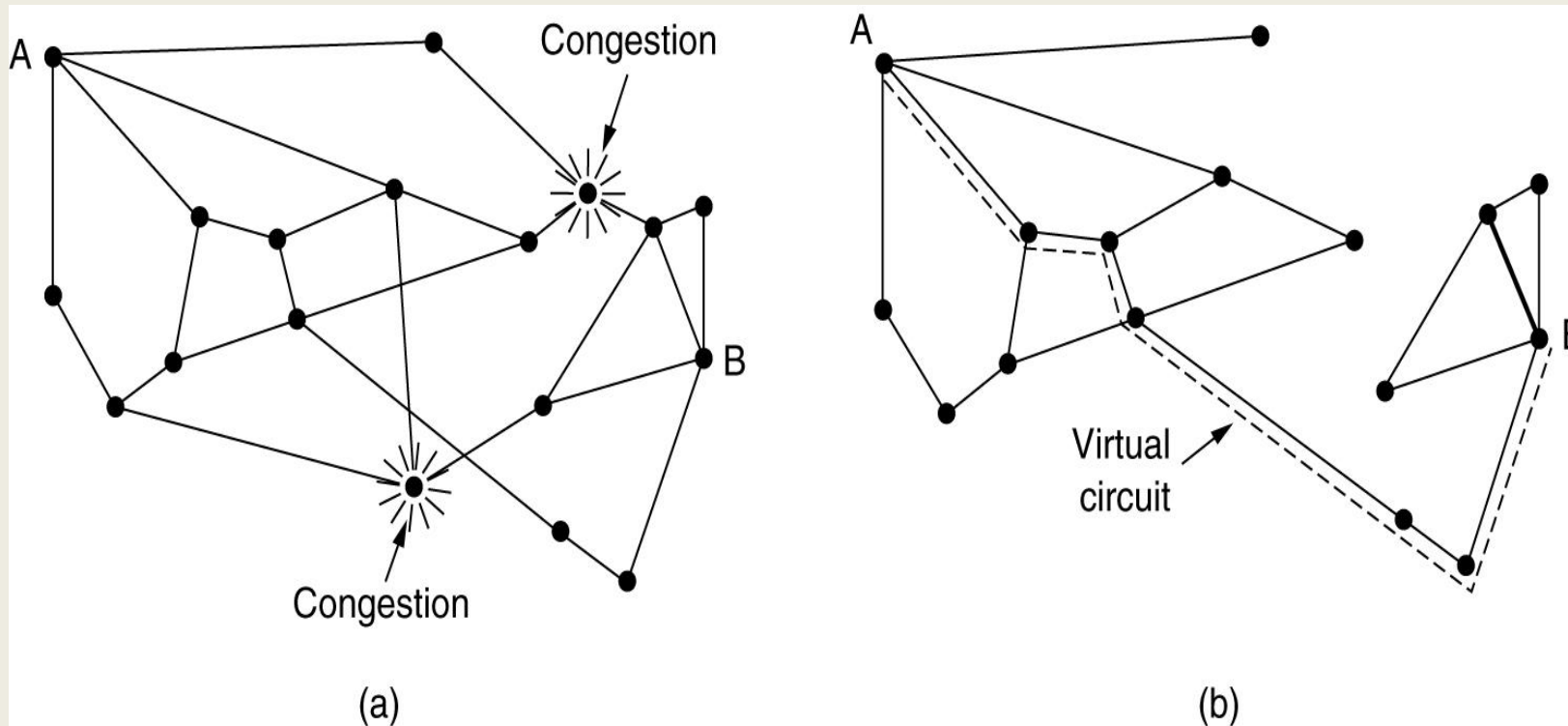
Admission Control

- One technique that is widely used in virtual-circuit networks to keep congestion at bay is **admission control**.
- The idea is simple: do not set up a new virtual circuit unless the network can carry the added traffic without becoming congested. Thus, attempts to set up a virtual circuit may fail.
- This is better than the alternative, as letting more people in when the network is busy just makes matters worse.
- However, virtual circuits in computer networks come in all shapes and sizes. Thus, the circuit must come with some characterization of its traffic if we are to apply admission control.

Admission Control

- Traffic is often described in terms of its rate and shape. The problem of how to describe it in a simple yet meaningful way is difficult because traffic is typically bursty—the average rate is only half the story.
- Equipped with traffic descriptions, the network can decide whether to admit the new virtual circuit. One possibility is for the network to reserve enough capacity along the paths of each of its virtual circuits that congestion will not occur. In this case, the traffic description is a service agreement for what the network will guarantee its users.

Congestion Control in Virtual-Circuit Subnets



(a) A congested subnet.

(b) A redrawn subnet, eliminates congestion and a virtual circuit from A to B.

Admission Control

- Admission control can also be combined with traffic-aware routing by considering routes around traffic hotspots as part of the setup procedure.
- Suppose that a host attached to router *A* wants to set up a connection to a host attached to router *B*. Normally, this connection would pass through one of the congested routers. To avoid this situation, we can redraw the network omitting the congested routers and all of their lines. The dashed line shows a possible route for the virtual circuit that avoids the congested routers.

Load Shedding

- When none of the above methods make the congestion disappear, routers can bring out the heavy artillery: load shedding.
- **Load shedding** is a fancy way of saying that when routers are being swamped by packets that they cannot handle, they just throw them away.
- The key question for a router drowning in packets is which packets to drop.
- The preferred choice may depend on the type of applications that use the network.
- For a file transfer, an old packet is worth more than a new one. This is because dropping packet 6 and keeping packets 7 through 10, for example, will only force the receiver to do more work to buffer data that it cannot yet use.

- In contrast, for real-time media, a new packet is worth more than an old one. This is because packets become useless if they are delayed and miss the time at which they must be played out to the user.
- The former policy (old is better than new) is often called **wine** and the latter (new is better than old) is often called **milk**.
- More intelligent load shedding requires cooperation from the senders. An example is packets that carry routing information. These packets are more important than regular data packets because they establish routes; if they are lost, the network may lose connectivity.

- Another example is that algorithms for compressing video, like MPEG, periodically transmit an entire frame and then send subsequent frames as differences from the last full frame. In this case, dropping a packet that is part of a difference is preferable to dropping one that is part of a full frame because future packets depend on the full frame.
- To implement an intelligent discard policy, applications must mark their packets to indicate to the network how important they are.
- Then, when packets have to be discarded, routers can first drop packets from the least important class, then the next most important class, and so on.
- Unless there is some significant incentive to avoid mark every packet as VERY IMPORTANT—NEVER, EVER DISCARD.

Traffic shaping:

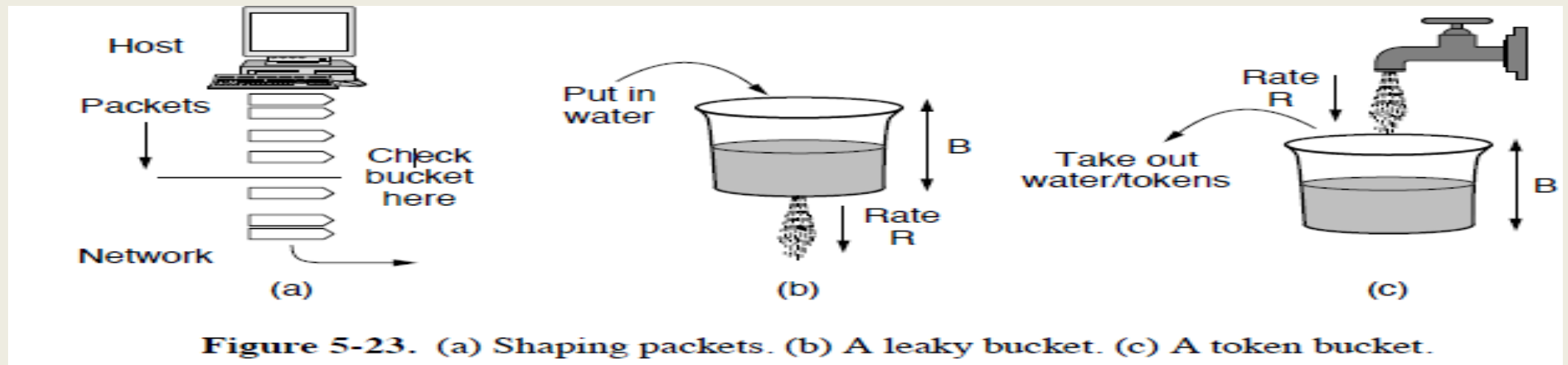
- Before the network can make performance guarantees, it must know what traffic is being guaranteed.
- Traffic shaping is a technique for regulating the average rate and burstiness of a flow of data that enters the network. The goal is to allow applications to transmit a wide variety of traffic that suits their needs, including some bursts, yet have a simple and useful way to describe the possible traffic patterns to the network.
- When a flow is set up, the user and the network (i.e., the customer and the provider) agree on a certain traffic pattern (i.e., shape) for that flow.

Traffic shaping:

- Sometimes this agreement is called an SLA (Service Level Agreement), especially when it is made over aggregate flows and long periods of time, such as all of the traffic for a given customer.
- Traffic shaping reduces congestion and thus helps the network live up to its promise.
- Packets in excess of the agreed pattern might be dropped by the network, or they might be marked as having lower priority. Monitoring a traffic flow is called traffic policing.

Traffic shaping:

- Now we will look at a more general way to characterize traffic, with the leaky bucket and token bucket algorithms.
- No matter the rate at which water enters the bucket, the outflow is at a constant rate, R , when there is any water in the bucket and zero when the bucket is empty. Also, once the bucket is full to capacity B , any additional water entering it spills over the sides and is lost.



Traffic shaping:

- Conceptually, each host is connected to the network by an interface containing a leaky bucket.
- To send a packet into the network, it must be possible to put more water into the bucket.
- If a packet arrives when the bucket is full, the packet must either be queued until enough water leaks out to hold it or be discarded.
- The former might happen at a host shaping its traffic for the network as part of the operating system. The latter might happen in hardware at a provider network interface that is policing traffic entering the network.
- This technique was proposed by Turner (1986) and is called the leaky bucket algorithm.

Traffic shaping:

- A different but equivalent formulation is to imagine the network interface as a bucket that is being filled.
- The tap is running at rate R and the bucket has a capacity of B , as before.
- Now to send a packet we must be able to take water, or tokens, as the contents are commonly called, out of the bucket (rather than putting water into the bucket).
- No more than a fixed number of tokens, B , can accumulate in the bucket, and if the bucket is empty, we must wait until more tokens arrive before we can send another packet.
- This algorithm is called the token bucket algorithm.

Active Queue Management:

- When congestion is imminent, it must tell the senders to throttle back their transmissions and slow down.
- Now look at some approaches to throttling traffic that can be used in both datagram networks and virtual-circuit networks.
- Each approach must solve two problems.
- First, routers must determine when congestion is approaching ideally before it has arrived.
- To do so, each router can continuously monitor the resources it is using. Three possibilities are the utilization of the output links, the buffering of queued packets inside the router, and the number of packets that are lost due to insufficient buffering.

Random Early Detection:

- Dealing with congestion when it first starts is more effective than letting it gum up the works and then trying to deal with it. This observation leads to an interesting twist on load shedding, which is to discard packets before all the buffer space is really exhausted.
- Idea is that most Internet hosts do not yet get congestion signals from routers in the form of ECN. Instead, the only reliable indication of congestion that hosts get from the network is packet loss. After all, it is difficult to build a router that does not drop packets when it is overloaded

- Transport protocols such as TCP are thus hardwired to react to loss as congestion, slowing down the source in response.
- The reasoning behind this logic is that TCP was designed for wired networks and wired networks are very reliable, so lost packets are mostly due to buffer overruns rather than transmission errors.
- This situation can be exploited to help reduce congestion. By having routers drop packets early, before the situation has become hopeless, there is time for the source to take action before it is too late. A popular algorithm for doing this is called **RED (Random Early Detection)**

- To determine when to start discarding, routers maintain a running average of their queue lengths.
- When the average queue length on some link exceeds a threshold, the link is said to be congested and a small fraction of the packets are dropped at random.
- Picking packets at random makes it more likely that the fastest senders will see a packet drop; this is the best option since the router cannot tell which source is causing the most trouble in a datagram network.
- The affected sender will notice the loss when there is no acknowledgement, and then the transport protocol will slow down.

Traffic Throttling

Choke Packets:

- The most direct way to notify a sender of congestion is to tell it directly.
- In this approach, the router selects a congested packet and sends a **choke packet** back to the source host, giving it the destination found in the packet.
- The original packet may be tagged (a header bit is turned on) so that it will not generate any more choke packets farther along the path and then forwarded in the usual way.
- To avoid increasing load on the network during a time of congestion, the router may only send choke packets at a low rate.

Traffic Throttling

- When the source host gets the choke packet, it is required to reduce the traffic sent to the specified destination, for example, by 50%.
- In a datagram network, simply picking packets at random when there is congestion is likely to cause choke packets to be sent to fast senders, because they will have the most packets in the queue.
- The feedback implicit in this protocol can help prevent congestion yet not throttle any sender unless it causes trouble.

Traffic Throttling

Explicit Congestion Notification:

- Instead of generating additional packets to warn of congestion, a router can tag any packet it forwards (by setting a bit in the packet's header) to signal that it is experiencing congestion.
- When the network delivers the packet, the destination can note that there is congestion and inform the sender when it sends a reply packet.
- The sender can then throttle its transmissions as before. This design is called **ECN (Explicit Congestion Notification)** and is used in the Internet.

Traffic Throttling

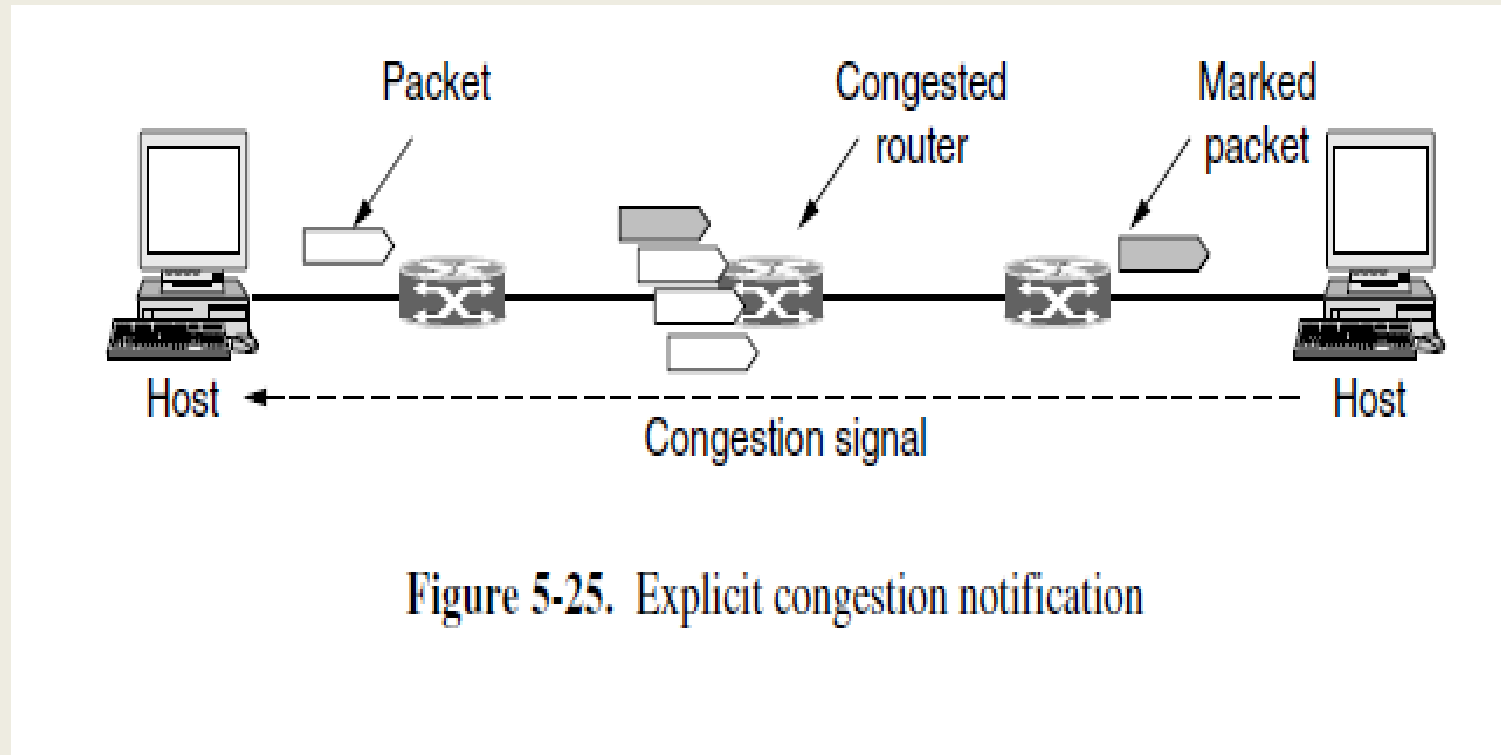


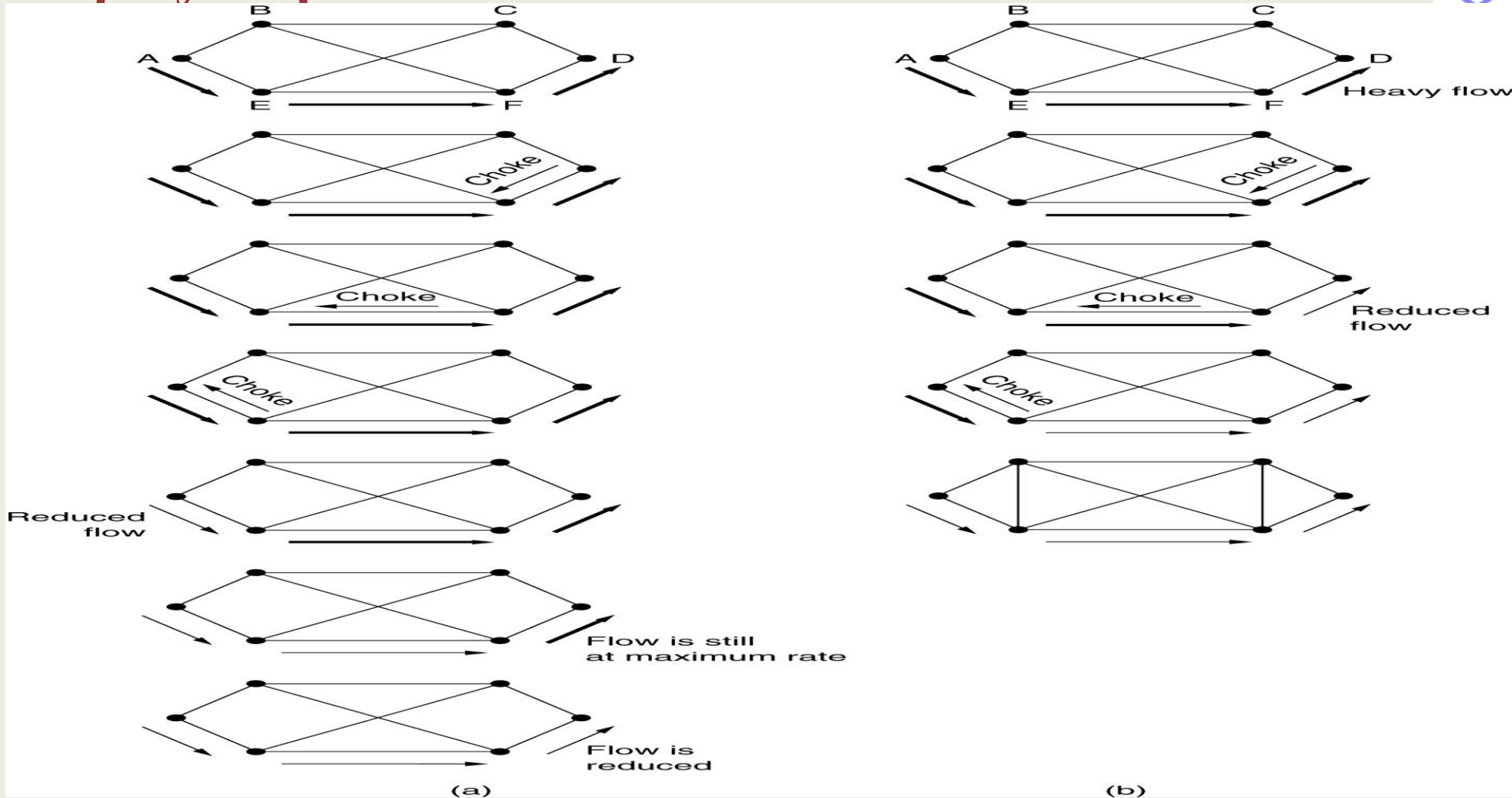
Figure 5-25. Explicit congestion notification

Traffic Throttling

Hop-by-Hop Backpressure:

- An alternative approach is to have the choke packet take effect at every hop it passes through.
- Here, as soon as the choke packet reaches F , F is required to reduce the flow to D . Doing so will require F to devote more buffers to the connection, since the source is still sending away at full blast, but it gives D immediate relief, like a headache remedy in a television commercial.
- In the next step, the choke packet reaches E , which tells E to reduce the flow to F . This action puts a greater demand on E 's buffers but gives F immediate relief. Finally, the choke packet reaches A and the flow genuinely slows down.

Hop-by-Hop Choke Packets



(a) A choke packet that affects only the source.

(b) A choke packet that affects each hop it passes through.

Quality of Service and Application QoE

- Each flow can be characterized by four primary parameters: bandwidth, delay, jitter, and loss.
- Bandwidth: Maximum data transfer rate of a network link (bits per second).
- Delay (Latency): Time taken for data to travel from source to destination.
- Jitter: Variation in packet arrival time (stability).
- Packet Loss: Percentage of packets failing to reach their destination.
- Together, these determine the QoS (Quality of Service) the flow requires.

Application	Bandwidth	Delay	Jitter	Loss
Email	Low	Low	Low	Medium
File sharing	High	Low	Low	Medium
Web access	Medium	Medium	Low	Medium
Remote login	Low	Medium	Medium	Medium
Audio on demand	Low	Low	High	Low
Video on demand	High	Low	High	Low
Telephony	Low	High	High	Low
Videoconferencing	High	High	High	Low

Figure 5-27. Stringency of applications' quality-of-service requirements.

Overprovisioning:

- An easy solution to provide good quality of service is to build a network with enough capacity for whatever traffic will be thrown at it.
- The name for this solution is overprovisioning.
- The resulting network will carry application traffic without significant loss and, assuming a decent routing scheme, will deliver packets with low latency.
- The trouble with this solution is that it is expensive.
- Four issues must be addressed to ensure quality of service:
 1. What applications need from the network.
 2. How to regulate the traffic that enters the network.
 3. How to reserve resources at routers to guarantee performance.
 4. Whether the network can safely accept more traffic.

Packet Scheduling:

- Algorithms that allocate router resources among the packets of a flow and between competing flows are called packet scheduling algorithms.
- Three different kinds of resources can potentially be reserved for different flows:
 1. Bandwidth.
 2. Buffer space.
 3. CPU cycles

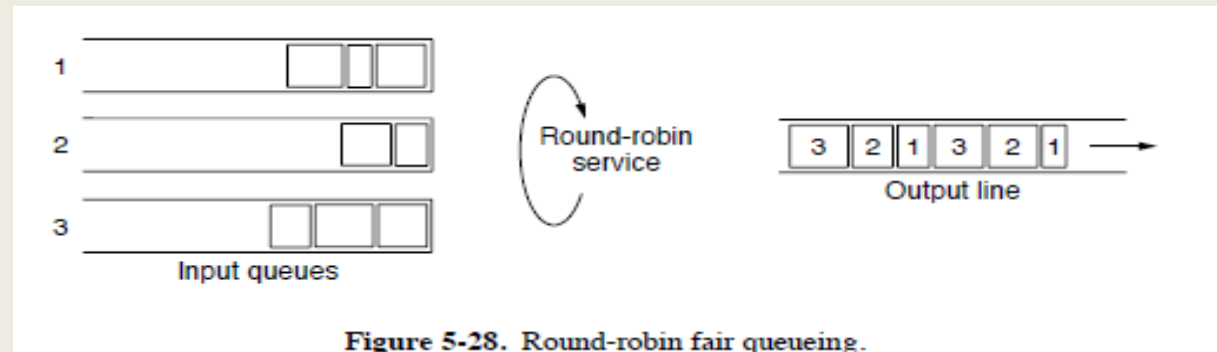
First-In First-Out (FIFO) Scheduling:

- Each router buffers packets in a queue for each output line until they can be sent, and they are sent in the same order that they arrived.
- This algorithm is known as FIFO (First-In First-Out), or equivalently FCFS (First-Come First-Served).
- FIFO routers usually drop newly arriving packets when the queue is full. Since the newly arrived packet would have been placed at the end of the queue, this behavior is called tail drop

Packet Scheduling:

Fair Queueing:

- FIFO scheduling is simple to implement, but it is not suited to providing good quality of service because when there are multiple flows, one flow can easily affect the performance of the other flows.
- Many packet scheduling algorithms have been devised that provide stronger isolation between flows and thwart attempts at interference.
- One of the first ones was the **fair queueing algorithm** where routers have separate queues, one for each flow for a given output line. When the line becomes idle, the router scans the queues round robin.
- It then takes the first packet on the next queue. In this way, with n hosts competing for the output line, each host gets to send one out of every n packets.



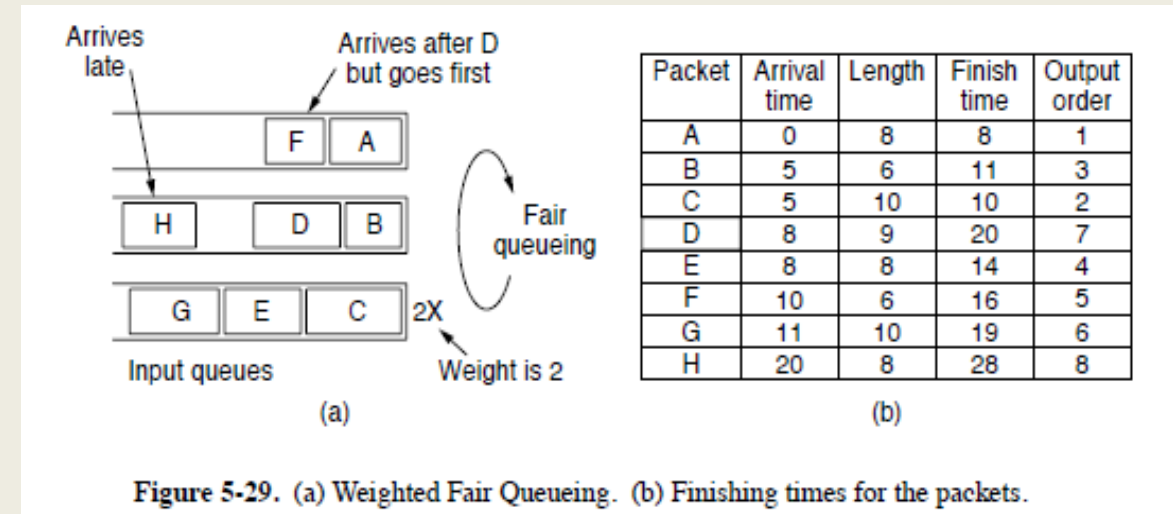
Packet Scheduling:

Fair Queueing:

- The algorithm has a flaw: it gives more bandwidth to hosts that use large packets than to hosts that use small packets.
- Demers et al. (1990) suggested an improvement in which the round robin is done in such a way as to simulate a byte-by-byte round robin, instead of a packet-by-packet round robin.
- The packets are then sorted in order of their finishing times and sent in that order.

Weighted Fair Queueing:

- One shortcoming of this algorithm in practice is that it gives all hosts the same priority.
- In many situations, it is desirable to give, for example, video servers more bandwidth than, say, file servers. This is easily possible by giving the video server two or more bytes per round.



Internetworks:

- It would be much simpler to join networks together if everyone used a single networking technology.
- The Internet is the prime example of this interconnection.
- The purpose of joining all these networks is to allow users on any of them to communicate with users on all the other ones.
- When you pay an ISP for Internet service, you may be charged depending on the bandwidth of your line, but what you are really paying for is the ability to exchange packets with any other host that is also connected to the Internet.
- Since networks often differ in important ways, getting packets from one network to another is not always so easy.
- We must address problems of heterogeneity, and also problems of scale as the resulting internet grows very large.

How Networks Differ:

Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering	Ordered and unordered delivery
Quality of service	Present or absent; many different kinds
Reliability	Different levels of loss
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, packet, byte, or not at all

Figure 5-36. Some of the many ways networks can differ.

Connecting Heterogeneous Networks:

- There are two basic choices for connecting different networks: we can build devices that translate or convert packets from each kind of network into packets for each other network, or as computer scientists often do, we can try to solve the problem by adding a layer of indirection and building a common layer on top of the different networks.
- IP provides a universal packet format that all routers recognize and that can be passed through almost every network. IP has extended its reach from computer networks to take over the telephone network.

Connecting Heterogeneous Networks:

- The first boundary the packet reaches is when it transitions from an 802.11 network to an MPLS network.
- Remember, 802.11 provides a connectionless service, but MPLS provides a connection-oriented service.
- This means that a virtual circuit must be set up to cross that network.
- Once the packet has traveled along the virtual circuit, it will reach the Ethernet network.
- At this boundary, the packet may be too large to be carried, since 802.11 can work with larger frames than Ethernet.
- To handle this problem, the packet is divided into fragments, and each fragment is sent separately. When the fragments reach the destination, they are reassembled.

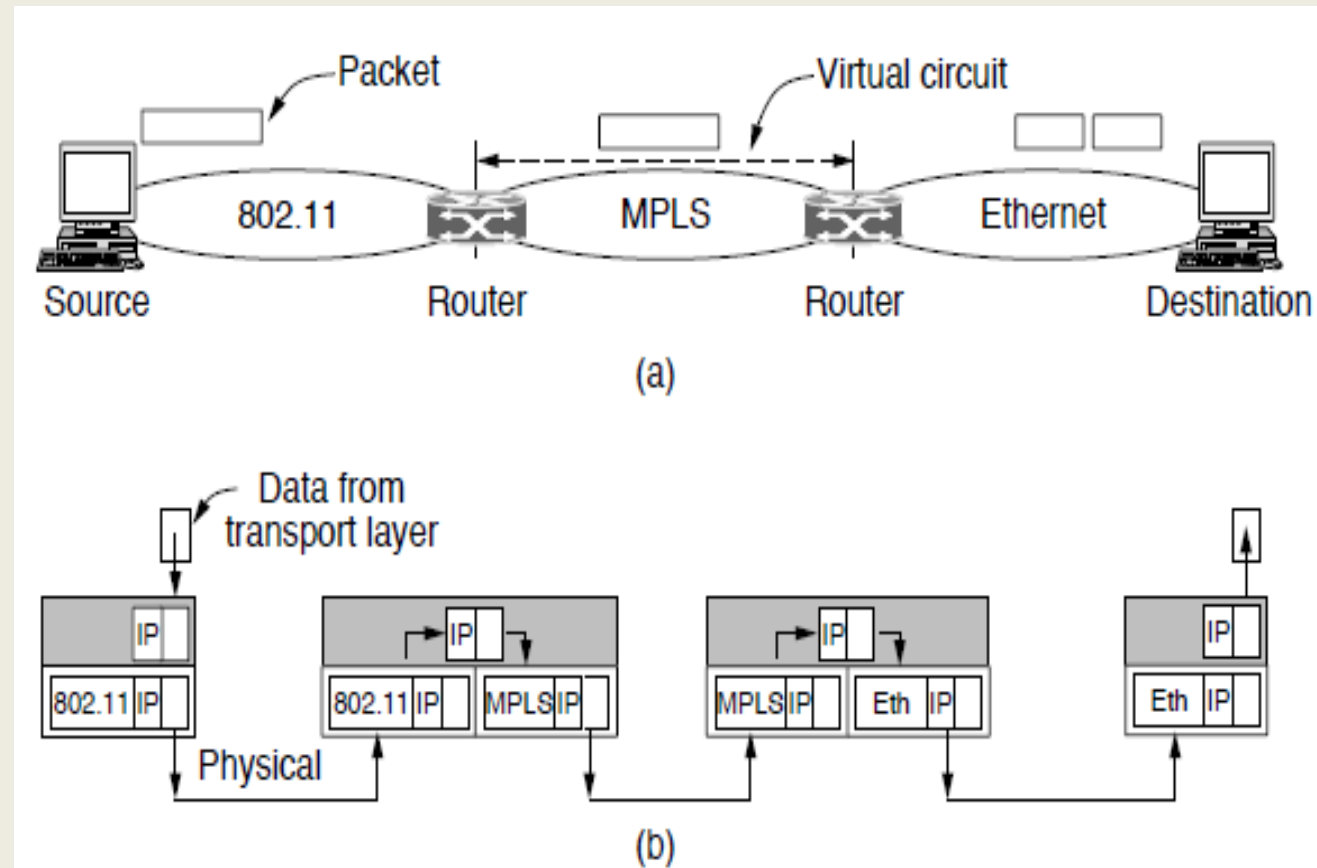


Figure 5-37. (a) A packet crossing different networks. (b) Network and link layer protocol processing.

Connecting Heterogeneous Networks:

- The source accepts data from the transport layer and generates a packet with the common network layer header, which is IP in this example.
- The network header contains the ultimate destination address, which is used to determine that the packet should be sent via the first router.
- So the packet is encapsulated in an 802.11 frame whose destination is the first router and transmitted. At the router, the packet is removed from the frame's data field and the 802.11 frame header is discarded.
- The router now examines the IP address in the packet and looks up this address in its routing table. Based on this address, it decides to send the packet to the second router next.
- When a packet is too long to be sent over Ethernet, it is split into two portions.
- Each of these portions is put into the data field of an Ethernet frame and sent to the Ethernet address of the destination.
- At the destination, the Ethernet header is stripped from each of the frames, and the contents are reassembled.

Connecting Endpoints Across Heterogeneous Networks:

- To send an IP packet to a host in the London office, a host in the Paris office constructs the packet containing an IPv6 address in London, and sends it to the multiprotocol router that connects the Paris IPv6 network to the IPv4 Internet.
- When this router gets the IPv6 packet, it encapsulates the packet with an IPv4 header addressed to the IPv4 side of the multiprotocol router that connects to the London IPv6 network.
- That is, the router puts a (IPv6) packet inside a (IPv4) packet.
- When this wrapped packet arrives, the London router removes the original IPv6 packet and sends it onward to the destination host.

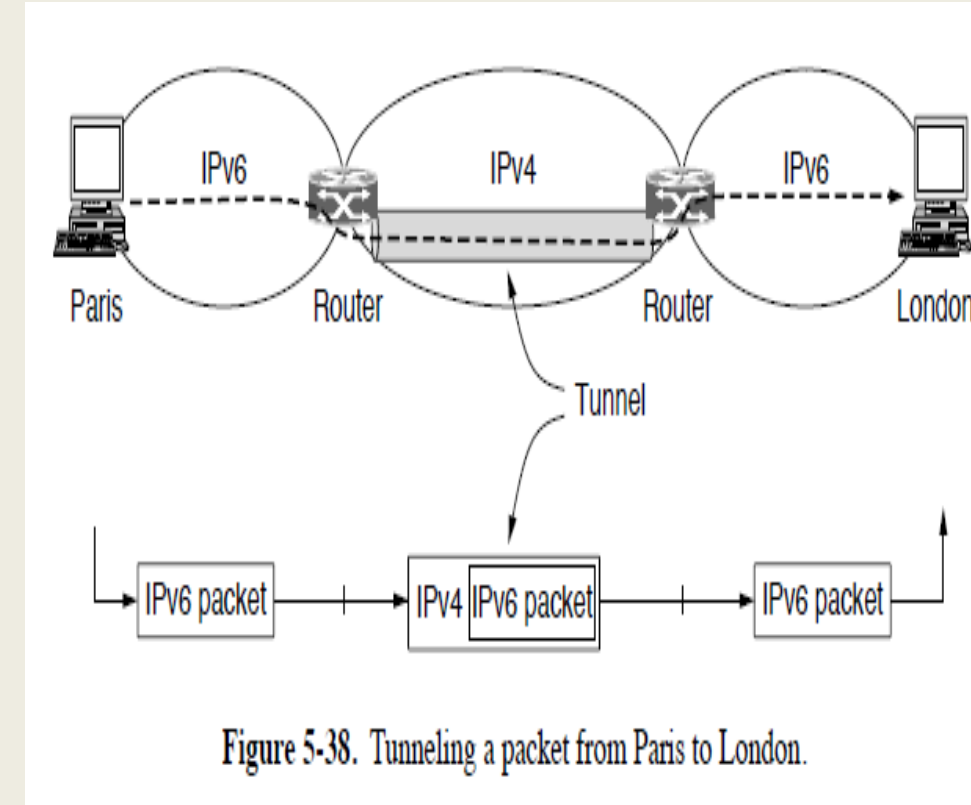


Figure 5-38. Tunneling a packet from Paris to London.

Internetwork Routing: Routing Across Multiple Networks

- Within each network, an intradomain or interior gateway protocol is used for routing. (“Gateway” is an older term for “router.”) It might be a link state protocol of the kind we have already described.
- Across the networks that make up the internet, an interdomain or exterior gateway protocol is used. The networks may all use different intradomain protocols, but they must use the same interdomain protocol.
- In the Internet, the interdomain routing protocol is called Border Gateway Protocol (BGP).
- In the Internet, a large determining factor is the business arrangements between ISPs. Each ISP may charge or receive money from the other ISPs for carrying traffic.
- Another factor is that if internetwork routing requires crossing international boundaries, various laws may suddenly come into play.
- All of these nontechnical factors are wrapped up in the concept of a routing policy that governs the way autonomous networks select the routes that they use.

Supporting Different Packet Sizes: Packet Fragmentation

- Each network or link imposes some maximum size on its packets.
- Hosts usually prefer to transmit large packets because this reduces packet overheads such as bandwidth wasted on header bytes.
- An obvious internetworking problem appears when a large packet wants to travel through a network whose maximum packet size is too small.
- A source does not usually know the path a packet will take through the network to a destination, so it certainly does not know how small a packet has to be to get there.
- This packet size is called the Path MTU (Path Maximum Transmission Unit).
- Even if the source did know the path MTU, packets are routed independently in a connectionless network such as the Internet.
- This routing means that paths may suddenly change, which can unexpectedly change the path MTU.

Supporting Different Packet Sizes: Packet Fragmentation

- The alternative solution to the problem is to allow routers to break up packets into fragments, sending each fragment as a separate network layer packet.
- Two opposing strategies exist for recombining the fragments back into the original packet. The first strategy is to make all the fragmentation caused by a “small-packet” network transparent to any subsequent networks through which the packet must pass on its way to the ultimate destination.
- The other fragmentation strategy is to refrain from recombining fragments at any intermediate routers. Once a packet has been fragmented, each fragment is treated as though it were an original packet.

Supporting Different Packet Sizes: Packet Fragmentation

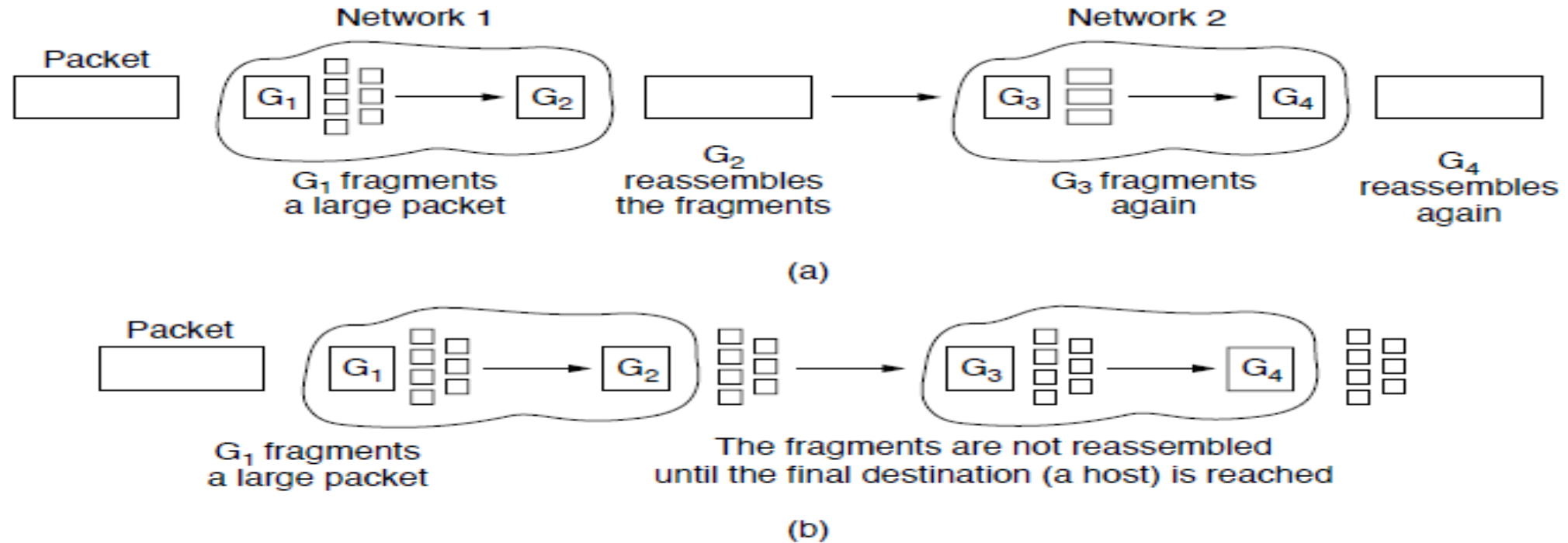
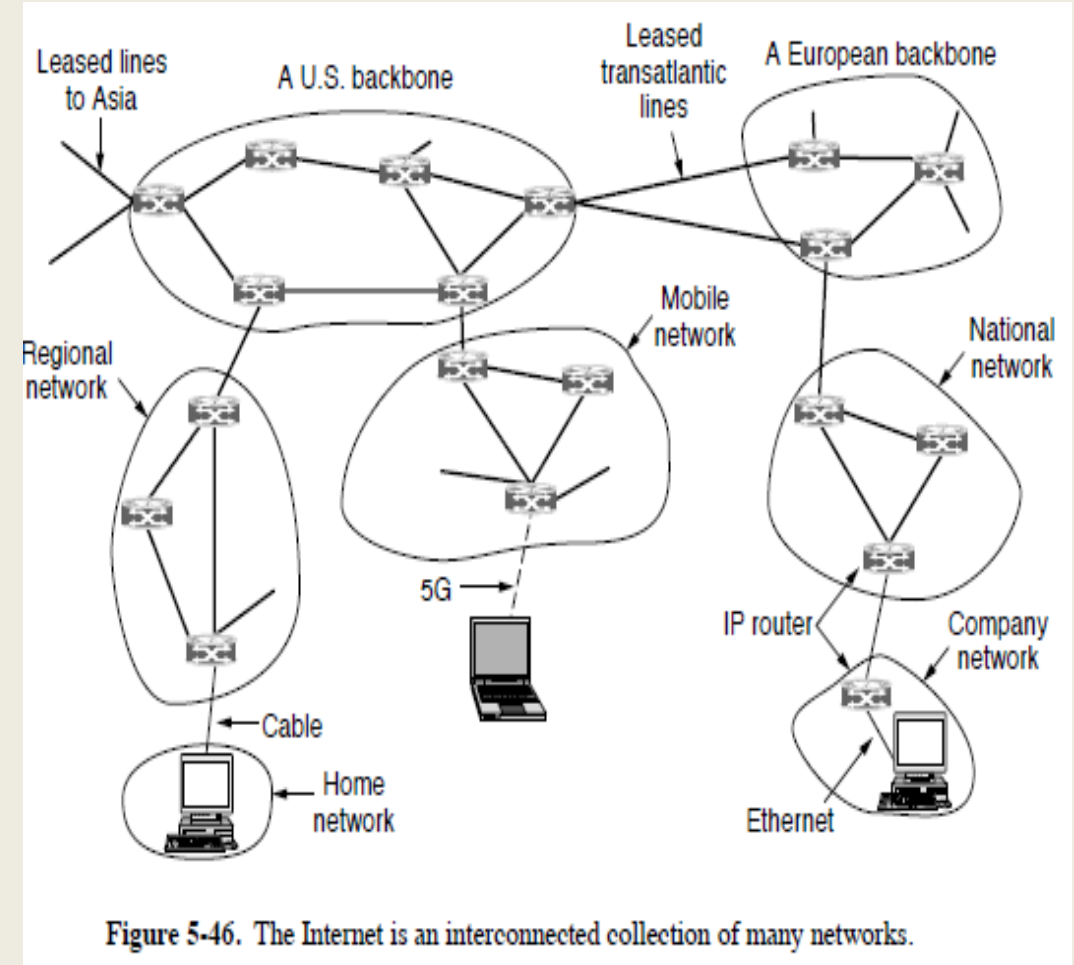


Figure 5-40. (a) Transparent fragmentation. (b) Nontransparent fragmentation.

The Network Layer in the Internet:

- In the network layer, the Internet can be viewed as a collection of networks or Autonomous Systems (ASes) that are interconnected.
- There is no real structure, but several major backbones exist.
- These are constructed from high-bandwidth lines and fast routers.
- The glue that holds the whole Internet together is the network layer protocol, IP (Internet Protocol).



The Network Layer in the Internet:

- In the network layer, the Internet can be viewed as a collection of networks or Autonomous Systems (ASes) that are interconnected.
- There is no real structure, but several major backbones exist.
- These are constructed from high-bandwidth lines and fast routers.
- The glue that holds the whole Internet together is the network layer protocol, IP (Internet Protocol).

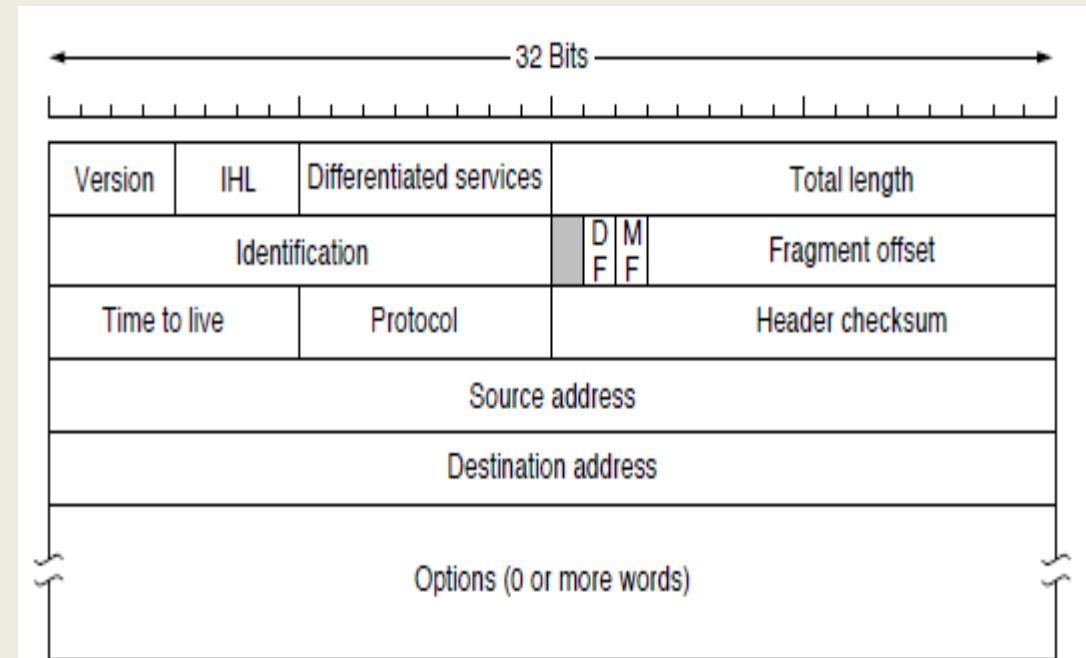


Figure 5-47. The IPv4 (Internet Protocol version 4) header.

The Network Layer in the Internet:

- An IPv4 datagram consists of a header part and a body or payload part.
- The header has a 20-byte fixed part and a variable-length optional part.
- The bits are transmitted from left to right and top to bottom, with the high-order bit of the Version field going first.
- The Version field keeps track of which version of the protocol the datagram belongs to.
- By including the version at the start of each datagram, it becomes possible to have a transition between versions over a long period of time.

The Network Layer in the Internet:

- Since the header length is not constant, a field in the header, IHL, is provided to tell how long the header is, in 32-bit words.
- The minimum value is 5, which applies when no options are present. The maximum value of this 4-bit field is 15, which limits the header to 60 bytes, and thus the Options field to 40 bytes.
- The Differentiated services field is one of the few fields that has changed its meaning (slightly) over the years. Originally, it was called the Type of service field.
- The Type of service field provided 3 bits to signal priority and 3 bits to signal whether a host cared more about delay, throughput, or reliability.

The Network Layer in the Internet:

- The Total length includes everything in the datagram—both header and data.
- The maximum length is 65,535 bytes.
- The Identification field is needed to allow the destination host to determine which packet a newly arrived fragment belongs to.
- All the fragments of a packet contain the same Identification value.
- DF stands for Don't Fragment. It is an order to the routers not to fragment the packet.
- MF stands for More Fragments.
- All fragments except the last one have this bit set. It is needed to know when all fragments of a datagram have arrived.

The Network Layer in the Internet:

- The Fragment offset tells where in the current packet this fragment belongs.
- All fragments except the last one in a datagram must be a multiple of 8 bytes—the elementary fragment unit.
- Since 13 bits are provided, there is a maximum of 8192 fragments per datagram, supporting a maximum packet length up to the limit of the Total length field.
- The TTL (Time to live) field is a counter used to limit packet lifetimes.
- It was originally supposed to count time in seconds, allowing a maximum lifetime of 255 sec.
- It must be decremented on each hop and is supposed to be decremented multiple times when a packet is queued for a long time in a router.

The Network Layer in the Internet:

- When it hits zero, the packet is discarded and a warning packet is sent back to the source host. This feature prevents packets from wandering around forever.
- The Protocol field tells it which transport process to give the packet to. TCP is one possibility, but so are UDP and some others. The numbering of protocols is global across the entire Internet.
- Since the header carries vital information such as addresses, it rates its own checksum for protection, the Header checksum.
- The Source address and Destination address indicate the IP address of the source and destination network interfaces.

The Network Layer in the Internet:

- The Options field was designed to provide an escape to allow subsequent versions of the protocol to include information not present in the original design.
- The Options field is padded out to a multiple of 4 bytes.

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

Figure 5-48. Some of the IP options.

The Network Layer in the Internet:

- IPv6 (IP version 6) uses 128-bit addresses.
- The Differentiated services field (originally called Traffic class) is used to distinguish the class of service for packets with different real-time delivery requirements.
- It is used with the differentiated service architecture for quality of service in the same manner as the field of the same name in the IPv4 packet.
- The Flow label field provides a way for a source and destination to mark groups of packets that have the same requirements and should be treated in the same way by the network, forming a pseudoconnection.

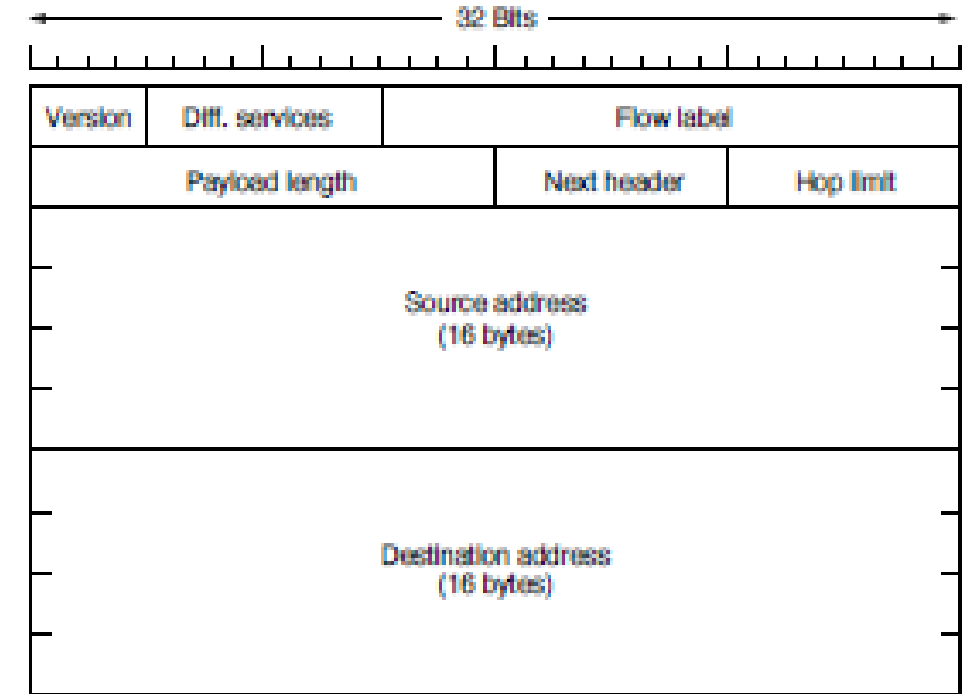


Figure 5-57. The IPv6 fixed header (required).

The Network Layer in the Internet:

- The Payload length field tells how many bytes follow the 40-byte header of Fig. 5-57.
- The 40 header bytes are no longer counted as part of the length (as they used to be).
- This change means the payload can now be 65,535 bytes instead of a mere 65,515 bytes.
- The Next header field lets the cat out of the bag.
- The reason the header could be simplified is that there can be additional (optional) extension headers.
- This field tells which of the (currently) six extension headers, if any, follow this one.
- The Hop limit field is used to keep packets from living forever, the same as the Time to live field in IPv4.

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

Figure 5-58. IPv6 extension headers.

Internet Control Message Protocol:

- When something unexpected occurs during packet processing at a router, the event is reported to the sender by the ICMP (Internet Control Message Protocol).
- ICMP is also used to test the Internet..
- Each ICMP message type is carried encapsulated in an IP packet.

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo and echo reply	Check if a machine is alive
Timestamp request/reply	Same as Echo, but with timestamp
Router advertisement/solicitation	Find a nearby router

Figure 5-61. The principal ICMP message types.